

Лабораторная работа №8

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Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

Выполнение лабораторной работы

Я продублировала в отчёте таблицы адресации (таблицы 8.1 и 8.2).

Таблица 8.1

Таблица адресов сетей

Устройства	Сеть IPv4	Сеть IPv6
PC1 – gw-01	10.0.10.0/24	2001:10::/64
PC2 – gw-03	10.0.11.0/24	2001:11::/64
gw-01 – gw-02	10.0.1.0/24	2001:1::/64
gw-02 – gw-03	10.0.2.0/24	2001:2::/64
gw-03 – gw-04	10.0.3.0/24	2001:3::/64
gw-04 – gw-01	10.0.4.0/24	2001:4::/64

Рис. 1: 8.1

Я продублировала в отчёте таблицы адресации (таблицы 8.1 и 8.2).

Таблица адресации				Таблица 8.2
Устройство	Интерфейс	Адрес IP/префикс	Шлюз по умолчанию	Следующее устройство
gw-01	eth0	10.0.10.1/24	n/a	PC1
gw-01	eth0	2001:10::1/64	n/a	PC1
gw-01	eth1	10.0.1.1/24	n/a	gw-02
gw-01	eth1	2001:1::1/64	n/a	gw-02
gw-01	eth2	10.0.4.2/24	n/a	gw-04
gw-01	eth2	2001:4::2/64	n/a	gw-04
gw-02	eth0	10.0.1.2/24	n/a	gw-01
gw-02	eth0	2001:1::2/64	n/a	gw-01
gw-02	eth1	10.0.2.1/24	n/a	gw-03
gw-02	eth1	2001:2::1/64	n/a	gw-03
gw-03	eth0	10.0.11.1/24	n/a	PC2
gw-03	eth0	2001:11::1/64	n/a	PC2
gw-03	eth1	10.0.2.2/24	n/a	gw-02
gw-03	eth1	2001:2::2/64	n/a	gw-02
gw-03	eth2	10.0.3.1/24	n/a	gw-04
gw-03	eth2	2001:3::1/64	n/a	gw-04
gw-04	eth0	10.0.3.2/24	n/a	gw-03
gw-04	eth0	2001:3::2/64	n/a	gw-03
gw-04	eth1	10.0.4.1/24	n/a	gw-01
gw-04	eth1	2001:4::1/64	n/a	gw-01
PC1	NIC	10.0.10.10/24	10.0.10.1	gw-01
PC1	NIC	2001:10::a/64	n/a	gw-01
PC2	NIC	10.0.11.10/24	10.0.11.1	gw-03
PC2	NIC	2001:11::a/64	n/a	gw-03

Рис. 2: 8.2

В графическом редакторе я схематично изобразила моделируемую сеть, ориентируясь на данные из таблиц адресации. Эти схемы я добавила в отчёт.

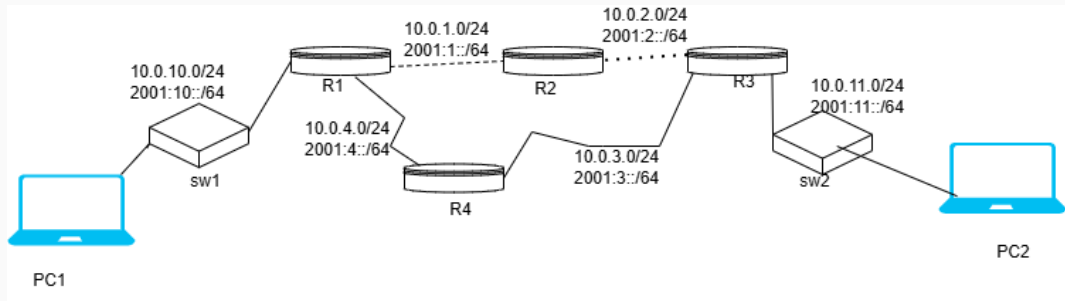


Рис. 3: схема

Построение и базовая настройка топологии

2.1 Создание проекта и размещение устройств

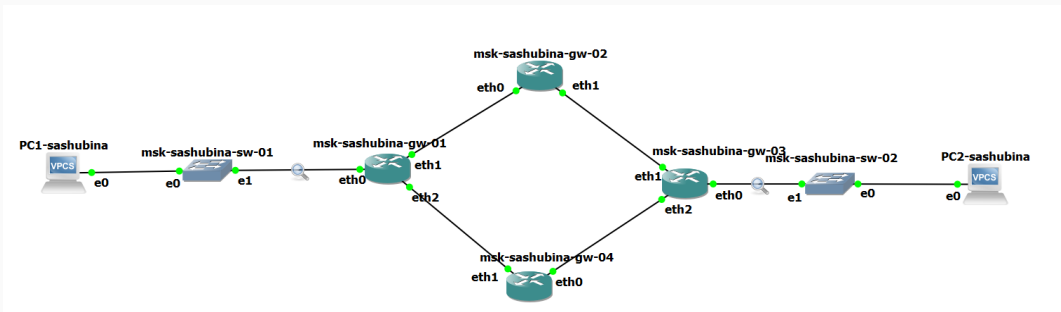


Рис. 4: новый проект

2.2 Настройка IPv4-адресации

```
VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip
NAME       : VPCS[1]
IP/MASK     : 10.0.10.10/24
GATEWAY     : 10.0.10.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20064
RHOST:PORT  : 127.0.0.1:20065
MTU         : 1500

VPCS> █
```

Рис. 5: PC1

```
VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.11.10/24
GATEWAY     : 10.0.11.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20082
RHOST:PORT  : 127.0.0.1:20083
MTU         : 1500

VPCS> █
```

Рис. 6: PC2

6. Я настроила IPv4-адреса на интерфейсах всех маршрутизаторов.

```
frf# configure terminal
frf(config)# hostname mak-sashubina-gw-01
mak-sashubina-gw-01(config)# interface eth0
mak-sashubina-gw-01(config-if)# ip address 10.0.10.1/24
mak-sashubina-gw-01(config-if)# no shutdown
mak-sashubina-gw-01(config-if)# exit
mak-sashubina-gw-01(config)# interface eth1
mak-sashubina-gw-01(config-if)# ip address 10.0.1.1/24
mak-sashubina-gw-01(config-if)# no shutdown
mak-sashubina-gw-01(config-if)# exit
mak-sashubina-gw-01(config)# interface eth2
mak-sashubina-gw-01(config-if)# ip address 10.0.4.2/24
mak-sashubina-gw-01(config-if)# no shutdown
mak-sashubina-gw-01(config-if)# exit
mak-sashubina-gw-01(config)# exit
mak-sashubina-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frf/frf.conf
[OK]
mak-sashubina-gw-01# show running-config
Building configuration...

Current configuration:
!
frf version 8.2.2
frf defaults traditional
hostname frf
hostname mak-sashubina-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 exit
!
interface eth1
 ip address 10.0.1.1/24
 exit
!
interface eth2
 ip address 10.0.4.2/24
 exit
!
end
mak-sashubina-gw-01#
```

Рис. 7: gw-01

Настройка для msk-sashubina-gw-02:

```
frr# configure terminal
frr(config)# hostname msk-sashubina-gw-02
msk-sashubina-gw-02(config)# interface eth0
msk-sashubina-gw-02(config-if)# ip address 10.0.1.2/24
msk-sashubina-gw-02(config-if)# no shutdown
msk-sashubina-gw-02(config-if)# exit
msk-sashubina-gw-02(config)# interface eth1
msk-sashubina-gw-02(config-if)# ip address 10.0.2.1/24
msk-sashubina-gw-02(config-if)# no shutdown
msk-sashubina-gw-02(config-if)# exit
msk-sashubina-gw-02(config)# exit
msk-sashubina-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-sashubina-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 exit
!
interface eth1
 ip address 10.0.2.1/24
 exit
!
end
msk-sashubina-gw-02#
```

Рис. 8: gw-02

Настройка для msk-sashubina-gw-03:

```
frf# configure terminal
frf(config)# hostname msk-sashubina-gw-03
msk-sashubina-gw-03(config)# interface eth0
msk-sashubina-gw-03(config-if)# ip address 10.0.11.1/24
msk-sashubina-gw-03(config-if)# no shutdown
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# interface eth1
msk-sashubina-gw-03(config-if)# ip address 10.0.2.2/24
msk-sashubina-gw-03(config-if)# no shutdown
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# interface eth2
msk-sashubina-gw-03(config-if)# ip address 10.0.3.1/24
msk-sashubina-gw-03(config-if)# no shutdown
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# exit
msk-sashubina-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frf/frf.conf
[OK]
msk-sashubina-gw-03# show running-config
Building configuration...

Current configuration:
!
frf version 8.2.2
frf defaults traditional
hostname frf
hostname msk-sashubina-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-sashubina-gw-03#
```

Рис. 9: gw-03

Настройка для msk-sashubina-gw-04:

```
frr# configure terminal
frr(config)# hostname msk-sashubina-gw-04
msk-sashubina-gw-04(config)# interface eth0
msk-sashubina-gw-04(config-if)# ip address 10.0.3.2/24
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)# exit
msk-sashubina-gw-04(config)# interface eth1
msk-sashubina-gw-04(config-if)# ip address 10.0.4.1/24
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)# exit
msk-sashubina-gw-04(config)# exit
msk-sashubina-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-sashubina-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-sashubina-gw-04#
```

Рис. 10: gw-04

7. Я присвоила IPv6-адреса оконечным устройствам.

```
VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 20064
RHOST:PORT          : 127.0.0.1:20065
MTU                 : 1500

VPCS> █
```

Рис. 11: PC1

```
VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 20082
RHOST:PORT           : 127.0.0.1:20083
MTU                 : 1500

VPCS> █
```

Рис. 12: PC2

8. Я настроила IPv6-адреса на интерфейсах маршрутизаторов.

```
mek-sashubina-gw-01# configure terminal
mek-sashubina-gw-01(config)# ipv6 forwarding
mek-sashubina-gw-01(config)# interface eth0
mek-sashubina-gw-01(config-if)# ipv6 address 2001:10::1/64
mek-sashubina-gw-01(config-if)# no ipv6 nd suppress-ra
mek-sashubina-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
mek-sashubina-gw-01(config-if)# no shutdown
mek-sashubina-gw-01(config-if)# exit
mek-sashubina-gw-01(config)# interface eth1
mek-sashubina-gw-01(config-if)# ipv6 address 2001:11:1/64
mek-sashubina-gw-01(config-if)# no shutdown
mek-sashubina-gw-01(config-if)# exit
mek-sashubina-gw-01(config)# interface eth2
mek-sashubina-gw-01(config-if)# ipv6 address 2001:4::2/64
mek-sashubina-gw-01(config-if)# no shutdown
mek-sashubina-gw-01(config-if)# exit
mek-sashubina-gw-01(config)# exit
mek-sashubina-gw-01# write memory
Note: this version of vtysh newer writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
mek-sashubina-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname mek-sashubina-gw-01
service integrated-vtysh-conf
!
interface eth0
 ip address 10.0.10.1/24
 ipv6 address 2001:10::1/64
 ipv6 nd prefix 2001:10::/64
 no ipv6 nd suppress-ra
 exit
!
interface eth1
 ip address 10.0.1.1/24
 ipv6 address 2001:11:1/64
 exit
!
interface eth2
 ip address 10.0.4.2/24
 ipv6 address 2001:4::2/64
 exit
```

Рис. 13: gw-01

Настройка для msk-sashubina-gw-02:

```
msk-sashubina-gw-02# configure terminal
msk-sashubina-gw-02(config)# ipv6 forwarding
msk-sashubina-gw-02(config)# interface eth0
msk-sashubina-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-sashubina-gw-02(config-if)# no shutdown
msk-sashubina-gw-02(config-if)# exit
msk-sashubina-gw-02(config)# interface eth1
msk-sashubina-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-sashubina-gw-02(config-if)# no shutdown
msk-sashubina-gw-02(config-if)# exit
msk-sashubina-gw-02(config)# exit
msk-sashubina-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/ffr/frf.conf
[OK]
msk-sashubina-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-sashubina-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ip address 10.0.2.1/24
 ipv6 address 2001:2::1/64
exit
!
end
msk-sashubina-gw-02#
```

Рис. 14: gw-02

Настройка для msk-sashubina-gw-03:

```
msk-sashubina-gw-03# configure terminal
msk-sashubina-gw-03(config)# ipv6 forwarding
msk-sashubina-gw-03(config)# interface eth0
msk-sashubina-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-sashubina-gw-03(config-if)# no ipv6 nd suppress-ra
msk-sashubina-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-sashubina-gw-03(config-if)# no shutdown
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# interface eth1
msk-sashubina-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-sashubina-gw-03(config-if)# no shutdown
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# interface eth2
msk-sashubina-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-sashubina-gw-03(config-if)# no shutdown
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# exit
msk-sashubina-gw-03# write memory
Note: this version of vtysh saves vtysh.conf
Building configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 3.2.2
frr defaults traditional
hostname frr
hostname msk-sashubina-gw-03
service integrated-vtysh-conf
!
interface eth0
ip address 10.0.11.1/24
ipv6 address 2001:11::1/64
ipv6 nd prefix 2001:11::/64
no ipv6 nd suppress-ra
exit
!
interface eth1
ip address 10.0.2.2/24
ipv6 address 2001:2::2/64
exit
!
interface eth2
ip address 10.0.3.1/24
ipv6 address 2001:3::1/64
exit
```

Рис. 15: gw-03

Настройка для msk-sashubina-gw-04:

```
msk-sashubina-gw-04# configure terminal
msk-sashubina-gw-04(config)# ipv6 forwarding
msk-sashubina-gw-04(config)# interface eth0
msk-sashubina-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)# exit
msk-sashubina-gw-04(config)# interface eth1
msk-sashubina-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)# exit
msk-sashubina-gw-04(config)# exit
msk-sashubina-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 6.2.2
frr defaults traditional
hostname frr
hostname msk-sashubina-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
 exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
 exit
!
end
msk-sashubina-gw-04#
```

Рис. 16: gw-04

3.1 Настройка RIPv2 для IPv4

```
msk-sashubina-gw-01# configure terminal
msk-sashubina-gw-01(config)# router rip
msk-sashubina-gw-01(config-router)# version 2
msk-sashubina-gw-01(config-router)# network eth0
msk-sashubina-gw-01(config-router)# network eth1
msk-sashubina-gw-01(config-router)# network eth2
msk-sashubina-gw-01(config-router)# exit
msk-sashubina-gw-01(config)# exit
msk-sashubina-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-01#
```

Рис. 17: gw-01

Настройка для msk-sashubina-gw-02:

```
msk-sashubina-gw-02# configure terminal
msk-sashubina-gw-02(config)# router rip
msk-sashubina-gw-02(config-router)# version 2
msk-sashubina-gw-02(config-router)# network eth0
msk-sashubina-gw-02(config-router)# network eth1
msk-sashubina-gw-02(config-router)# exit
msk-sashubina-gw-02(config)# exit
msk-sashubina-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-02#
```

Рис. 18: gw-02

Настройка для msk-sashubina-gw-03:

```
msk-sashubina-gw-03# configure terminal
msk-sashubina-gw-03(config)# router rip
msk-sashubina-gw-03(config-router)# version 2
msk-sashubina-gw-03(config-router)# network eth0
msk-sashubina-gw-03(config-router)# network eth1
msk-sashubina-gw-03(config-router)# network eth2
msk-sashubina-gw-03(config-router)# exit
msk-sashubina-gw-03(config)# exit
msk-sashubina-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-03#
```

Рис. 19: gw-03

Настройка для msk-sashubina-gw-04:

```
msk-sashubina-gw-04# configure terminal
msk-sashubina-gw-04(config)# router rip
msk-sashubina-gw-04(config-router)# version 2
msk-sashubina-gw-04(config-router)# network eth0
msk-sashubina-gw-04(config-router)# network eth1
msk-sashubina-gw-04(config-router)# exit
msk-sashubina-gw-04(config)# exit
msk-sashubina-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-04#
```

Рис. 20: gw-04

2. Я убедилась, что маршрутизация по RIP работает, выполнив на каждом маршрутизаторе команды:

```
ask-nashubina-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, V - VNC, V - VNC-Direct, A - Appel, F - FHR,
        I - OpenFabric.
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

R* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:01:04
R* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:00:29
R* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:01:42
ask-nashubina-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
S&B-CODES:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

Network        Next Hop        Metric From        Tag Time
C(i) 10.0.1.0/24 0.0.0.0         1 self            0
R(m) 10.0.2.0/24 10.0.1.2        2 10.0.1.2         0 02:46
R(m) 10.0.3.0/24 10.0.4.1        2 10.0.4.1         0 02:52
C(i) 10.0.4.0/24 0.0.0.0         1 self            0
C(i) 10.0.10.0/24 0.0.0.0         1 self            0
R(m) 10.0.11.0/24 10.0.1.2        3 10.0.1.2         0 02:46
ask-nashubina-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 7 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface    Send Recv  Key-chain
    eth0         2      2
    eth1         2      2
    eth2         2      2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway     BadRackets BadRoutes  Distance Last Update
    10.0.1.2      0          0       120  00:00:21
    10.0.4.1      0          0       120  00:00:15
  Distance: (default is 120)
ask-nashubina-gw-01#
```

Рис. 21: gw-01

2. Я убедилась, что маршрутизация по RIP работает, выполнив на каждом маршрутизаторе команды

```
ask-sashubina-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, V - VNC, V - VNC-Direct, A - Aalal, F - FRR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

R>* 10.0.3.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:40
R>* 10.0.4.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:04:03
R>* 10.0.10.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:04:03
R>* 10.0.11.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:33
ask-sashubina-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
(n) - normal, (s) - static, (d) - default, (r) - redistribute,
(i) - interface

      Network      Next Hop      Metric From      Tag Time
C(i) 10.0.1.0/24    0.0.0.0        1 self          0
C(i) 10.0.2.0/24    0.0.0.0        1 self          0
R(n) 10.0.3.0/24    10.0.2.2        2 10.0.2.2        0 02:46
R(n) 10.0.4.0/24    10.0.1.1        2 10.0.1.1        0 02:52
R(n) 10.0.10.0/24   10.0.1.1        2 10.0.1.1        0 02:52
R(n) 10.0.11.0/24   10.0.2.2        2 10.0.2.2        0 02:46
ask-sashubina-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 4 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv   Key-chain
    eth0           2    2
    eth1           2    2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
    10.0.1.1      0          0        120 00:00:14
    10.0.2.2      0          0        120 00:00:21
  Distance: (default is 120)
ask-sashubina-gw-02#
```

Рис. 22: gw-02

2. Я убедилась, что маршрутизация по RIP работает, выполнив на каждом маршрутизаторе команды

```
ask-sashubina-gw-03# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, M - NHRP,
        T - Table, V - VNC, V - VNC-Direct, A - Acl, F - FRR,
        f - OpenFabric,
        > - Selected route, * - FIB route, q - queued, r - rejected, b - backup
        t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:03:26
R>* 10.0.4.0/24 [120/2] via 10.0.3.2, eth2, weight 1, 00:02:11
R>* 10.0.10.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:03:26
ask-sashubina-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (a) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface

Network        Next Hop        Metric From        Tag Time
R(m) 10.0.1.0/24 10.0.2.1        2 10.0.2.1        0 02:14
C(i) 10.0.2.0/24 0.0.0.0         1 self            0
C(i) 10.0.3.0/24 0.0.0.0         1 self            0
R(m) 10.0.4.0/24 10.0.3.2        2 10.0.3.2        0 02:45
R(m) 10.0.10.0/24 10.0.2.1        3 10.0.2.1        0 02:14
C(i) 10.0.11.0/24 0.0.0.0         1 self            0
ask-sashubina-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 8 seconds
  Timeout after 180 seconds, garbage collect after 180 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv  Key-chain
    eth0           2      2
    eth1           2      2
    eth2           2      2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
    10.0.2.1      0          0        120  00:00:02
    10.0.3.2      0          0        120  00:00:21
  Distance: (default is 120)
ask-sashubina-gw-03#
```

Рис. 23: gw-03

2. Я убедилась, что маршрутизация по RIP работает, выполнив на каждом маршрутизаторе команды

```
msk-sashubina-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, V - VNC, V - VNC-Direct, R - Rabel, F - FRR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:50
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:57
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:50
R>* 10.0.11.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:57
msk-sashubina-gw-04# show ip rip
Codes: R - RIP, C - connected, S - static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface

      Network      Next Hop      Metric From      Tag Time
R(n) 10.0.1.0/24    10.0.4.2       2 10.0.4.2       0 02:44
R(n) 10.0.2.0/24    10.0.3.1       2 10.0.3.1       0 02:34
C(i) 10.0.3.0/24    0.0.0.0        0 self           0
C(i) 10.0.4.0/24    0.0.0.0        0 self           0
R(n) 10.0.10.0/24   10.0.4.2       2 10.0.4.2       0 02:44
R(n) 10.0.11.0/24   10.0.3.1       2 10.0.3.1       0 02:34
msk-sashubina-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 28 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
  Interface      Send Recv  Key-chain
  eth0           2    2
  eth1           2    2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
  10.0.3.1        0          0       120  00:00:12
  10.0.4.2        0          0       120  00:00:13
  Distance: (default is 120)
msk-sashubina-gw-04#
```

Рис. 24: gw-04

3. Я проверила пути прохождения пакетов и устойчивость сети к сбоям.

```
PC1> ping 10.0.11.10
P: 84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=13.441 ms
P: 84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=4.725 ms
P: 84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=5.270 ms
P: 84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=3.791 ms
P: 84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.097 ms
```

Рис. 25: PC1 ping

3. Я проверила пути прохождения пакетов и устойчивость сети к сбоям.

```
VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    2.703 ms  0.377 ms  0.410 ms
 2  10.0.1.2     4.563 ms  1.659 ms  1.457 ms
 3  10.0.2.2     6.216 ms  2.637 ms  2.749 ms
 4  10.0.11.10   7.142 ms  2.692 ms  2.943 ms

VPCS> 
```

Рис. 26: PC1 trace

Проверила метрики RIP на msk-sashubina-gw-01:

```
msk-sashubina-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self               0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2            0 02:39
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1            0 02:46
C(i) 10.0.4.0/24       0.0.0.0           1 self               0
C(i) 10.0.10.0/24      0.0.0.0           1 self               0
R(n) 10.0.11.0/24      10.0.1.2          3 10.0.1.2            0 02:39
msk-sashubina-gw-01#
```

Рис. 27: gw-01

Отключила интерфейс eth0 на msk-sashubina-gw-04:

```
msk-sashubina-gw-02# configure terminal
msk-sashubina-gw-02(config)# interface eth0
msk-sashubina-gw-02(config-if)# shutdown
msk-sashubina-gw-02(config-if)#
```

Рис. 28: gw-02

Отключила интерфейс eth0 на msk-sashubina-gw-04:

```
msk-sashubina-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.4.1          3 10.0.4.1           0 02:29
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:29
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          3 10.0.4.1           0 02:29
msk-sashubina-gw-01#
```

Рис. 29: gw-02

Отключила интерфейс eth0 на msk-sashubina-gw-04:

```
msk-sashubina-gw-04# configure terminal
msk-sashubina-gw-04(config)# interface eth0
msk-sashubina-gw-04(config-if)# shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 30: gw-04

Отключила интерфейс eth0 на msk-sashubina-gw-04:

```
msk-sashubina-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.4.1          16 10.0.4.1         0 01:26
R(n) 10.0.3.0/24       10.0.4.1          16 10.0.4.1         0 01:26
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          16 10.0.4.1         0 01:26
msk-sashubina-gw-01#
```

Рис. 31: gw-04

Пропинговала с PC1 адрес PC2.

```
VPCS> ping 10.0.11.10

*10.0.10.1 icmp_seq=1 ttl=64 time=2.371 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=2 ttl=64 time=0.856 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=3 ttl=64 time=0.855 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=4 ttl=64 time=0.804 ms (ICMP type:3, code:0, Destination net
work unreachable)
10.0.11.10 icmp_seq=5 timeout

VPCS> █
```

Рис. 32: ping

Включила интерфейс обратно:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# interface eth0
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 33: gw-04

Также пропингуем:

```
VPCS> ping 10.0.11.10  
  
84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=5.306 ms  
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=6.733 ms  
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=3.044 ms  
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=3.546 ms  
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=3.556 ms  
  
VPCS> 
```

Рис. 34: ping

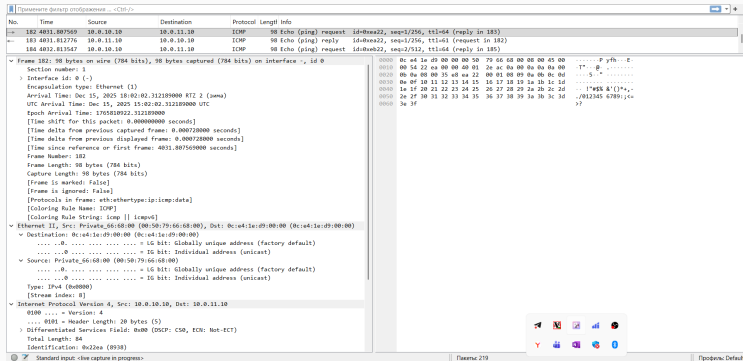


Рис. 35: захват трафика

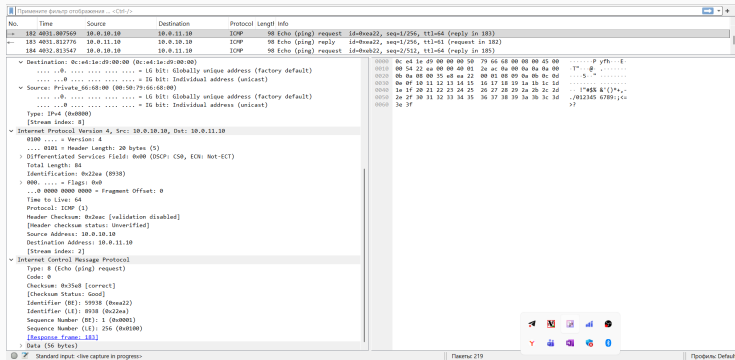


Рис. 36: захват трафика

Захват трафика

The screenshot displays the Wireshark interface with a packet capture of ICMP Echo (ping) traffic. The packet list on the left shows three packets: a request (No. 182), a reply (No. 183), and another request (No. 184). The selected packet (No. 183) is expanded, showing the Ethernet II header, Internet Protocol Version 4 header, and ICMP Echo (ping) reply details. The packet details pane on the right shows the raw packet data in hexadecimal and ASCII. The packet bytes pane at the bottom shows the packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
182	4811.1807569	10.0.11.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x022, seq=12356, ttl=64 (reply in 183)
183	4811.812776	10.0.11.10	10.0.11.10	ICMP	98	Echo (ping) reply id=0x022, seq=12356, ttl=64 (request in 182)
184	4812.813547	10.0.11.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x022, seq=2512, ttl=64 (reply in 185)

▼ Frame 183: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

- Section number: 1
- Interface id: 0 (-)
- Encapsulation type: Ethernet (1)
- Arrival Time: Dec 15, 2025 18:02:02.317396000 RTT: 2 (sma)
- UTC Arrival Time: Dec 15, 2025 15:02:02.317396000 UTC
- Epoch Arrival Time: 1765810922.317396000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 0.005207000 seconds]
- [Time delta from previous displayed frame: 0.005207000 seconds]
- [Time since reference or first frame: 4811.812776000 seconds]
- Frame Number: 183
- Frame Length: 98 bytes (784 bits)
- Capture Length: 98 bytes (784 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: ethertype:ip:icmp:data]
- [Coloring Rule Name: ICMP]
- [Coloring Rule String: icmp || icmpv6]
- ▼ Ethernet II, Src: 0c:04:1e:d9:00:00 (0c:04:1e:d9:00:00), Dst: Private_66:68:08 (08:50:70:66:68:08)
 - Destination: Private_66:68:08 (08:50:70:66:68:08)
 - ...0... = 16 bit: Globally unique address (factory default)
 - ...0... = 16 bit: Individual address (unicast)
 - Source: 0c:04:1e:d9:00:00 (0c:04:1e:d9:00:00)
 - ...0... = 16 bit: Globally unique address (factory default)
 - ...0... = 16 bit: Individual address (unicast)
 - Type: IPv4 (0x0008)
 - [Stream index: 8]
- ▼ Internet Protocol Version 4, Src: 10.0.11.10, Dst: 10.0.11.10
 - 0100 ... = Version: 4
 - ...0101 = Header Length: 20 bytes (5)
 - ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 84
 - Identification: 0x22ea (8938)

Standard input: <live capture in progress> | Память: 220 | Профайл: Def

Рис. 37: захват трафика

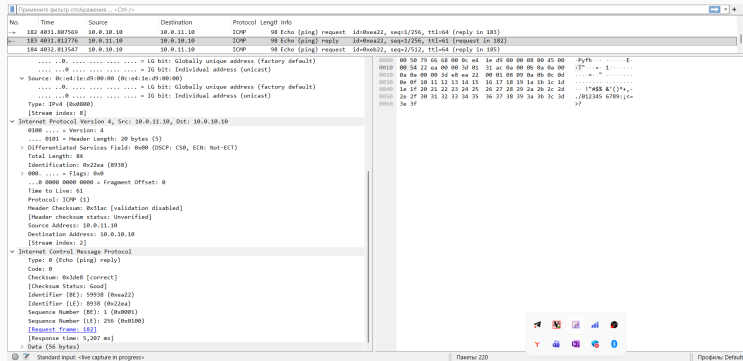


Рис. 38: захват трафика

No.	Time	Source	Destination	Protocol	Length	Info
193	4036.866113	Private_66:68:00	0c:e4:1e:d9:00:00	ARP	60	10.0.10.10 is at 00:50:79:66:68:00
194	4047.399510	10.0.10.1	224.0.0.9	RIPv2	146	Response
195	4079.414047	10.0.10.1	224.0.0.9	RIPv2	146	Response
196	4114.429650	10.0.10.1	224.0.0.9	RIPv2	146	Response
197	4139.435719	10.0.10.1	224.0.0.9	RIPv2	146	Response

▼ Frame 196: 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits) on interface -, id 0	0000	01 00 5e 00 00 09 0c e4 1e d9 00 00 00 00 45 c0	E.....
Section number: 1	0010	00 94 1e 04 40 00 01 11 66 92 0a 00 0a 01 a0 00	F.....
Interface id: 0 (-)	0020	00 09 02 08 02 08 00 70 c2 d9 02 02 00 00 00 02	p.....
Encapsulation type: Ethernet (1)	0030	00 00 0a 00 01 00 ff ff ff 00 00 00 00 00 00 00	X.....
Arrival Time: Dec 15, 2025 10:02:17.904130000 RTZ 2 (smba)	0040	00 01 00 02 00 00 0a 00 02 00 ff ff 00 00 00 00	
UTC Arrival Time: Dec 15, 2025 15:02:17.904130000 UTC	0050	00 00 00 00 02 00 02 00 00 0a 00 03 00 ff ff	
Epoch Arrival Time: 1705610937.904130000	0060	ff 00 00 00 00 00 00 00 02 00 02 00 00 0a 00	
[Time shift for this packet: 0.000000000 seconds]	0070	04 00 ff ff 00 00 00 00 00 00 00 00 01 00 02	
[Time delta from previous captured frame: 10.53397000 seconds]	0080	00 00 0a 00 0b 00 ff ff ff 00 00 00 00 00 00	
[Time delta from previous displayed frame: 10.53397000 seconds]	0090	00 03	..
[Time since reference or first frame: 4047.399510000 seconds]			
Frame Number: 194			
Frame Length: 146 bytes (1168 bits)			
Capture Length: 146 bytes (1168 bits)			
[Frame is marked: false]			
[Frame is ignored: False]			
[Protocols in frame: eth:ethertype:ip:udp:rip]			
[Coloring Rule Name: UDP]			
[Coloring Rule String: udp]			
▼ Ethernet II, Src: 0c:e4:1e:d9:00:00 (0c:e4:1e:d9:00:00), Dst: IP4multicast_09 (01:00:5e:00:00:09)			
Destination: IP4multicast_09 (01:00:5e:00:00:09)			
.....0..... = LG bit: Globally unique address (factory default)			
.....1..... = IG bit: Group address (multicast/broadcast)			
Source: 0c:e4:1e:d9:00:00 (0c:e4:1e:d9:00:00)			
.....0..... = LG bit: Globally unique address (factory default)			
.....0..... = IG bit: Individual address (unicast)			
Type: IP4 (0x0800)			
[Stream index: 6]			
▼ Internet Protocol Version 4, Src: 10.0.10.1, Dst: 224.0.0.9			
0100 = Version: 4			
.....0101 = Header Length: 20 bytes (5)			
Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)			

Рис. 39: захват трафика

Захват трафика

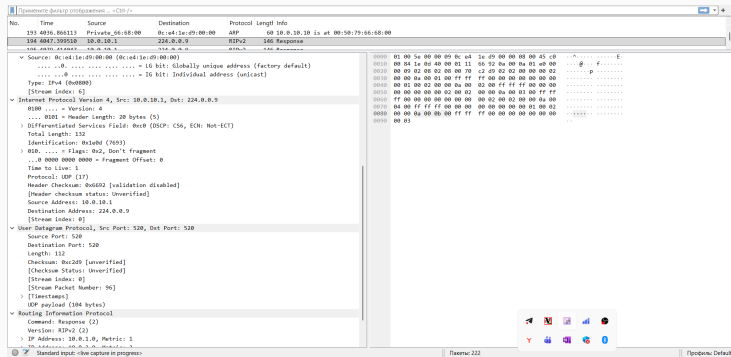


Рис. 40: захват трафика

3.2 Настройка RIPv6 для IPv6

```
msk-sashubina-gw-01# configure terminal
msk-sashubina-gw-01(config)# router ripng
msk-sashubina-gw-01(config-router)# network eth0
msk-sashubina-gw-01(config-router)# network eth1
msk-sashubina-gw-01(config-router)# network eth2
msk-sashubina-gw-01(config-router)# exit
msk-sashubina-gw-01(config)# exit
msk-sashubina-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-01# █
```

Рис. 41: gw-01

Настройка для msk-sashubina-gw-02:

```
msk-sashubina-gw-02(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-02(config-if)# router ripng
msk-sashubina-gw-02(config-router)# network eth0
msk-sashubina-gw-02(config-router)# network eth1
msk-sashubina-gw-02(config-router)# exit
msk-sashubina-gw-02(config)# exit
msk-sashubina-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-02#
```

Рис. 42: gw-02

Настройка для msk-sashubina-gw-03:

```
msk-sashubina-gw-03# configure terminal
msk-sashubina-gw-03(config)# router ripng
msk-sashubina-gw-03(config-router)# network eth0
msk-sashubina-gw-03(config-router)# network eth1
msk-sashubina-gw-03(config-router)# network eth2
msk-sashubina-gw-03(config-router)# exit
msk-sashubina-gw-03(config)# exit
msk-sashubina-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-03#
```

Рис. 43: gw-03

Настройка для msk-sashubina-gw-04:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# router ripng
msk-sashubina-gw-04(config-router)# network eth0
msk-sashubina-gw-04(config-router)# network eth1
msk-sashubina-gw-04(config-router)# exit
msk-sashubina-gw-04(config)# exit
msk-sashubina-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-04#
```

Рис. 44: gw-04

5. Я провела проверки для IPv6.

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=14.338 ms
2001:11::a icmp6_seq=2 ttl=58 time=7.529 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.373 ms
2001:11::a icmp6_seq=4 ttl=58 time=8.039 ms
2001:11::a icmp6_seq=5 ttl=58 time=7.385 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1   10.422 ms  2.818 ms  2.390 ms
 2 2001:1::2    10.024 ms  8.315 ms  5.651 ms
 3 2001:2::2    14.473 ms  13.475 ms 13.231 ms
 4 2001:11::a   9.710 ms  9.377 ms  8.628 ms

VPCS> █
```

Рис. 45: PC1 ping

Проверила метрики на msk-sashubina-gw-01:

```
msk-sashubina-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

   Network      Next Hop          Via      Metric Tag Time
C(i) 2001:1::/64      ::                self       1    0
R(n) 2001:2::/64      fe80::ec8:7fff:fe62:0    eth1       2    0 02:36
R(n) 2001:3::/64      fe80::e82:3aff:fec0:1    eth2       2    0 02:30
C(i) 2001:4::/64      ::                self       1    0
C(i) 2001:10::/64     ::                self       1    0
R(n) 2001:11::/64     fe80::ec8:7fff:fe62:0    eth1       3    0 02:36
msk-sashubina-gw-01#
```

Рис. 46: gw-01

Отключила интерфейс eth0 на msk-sashubina-gw-04

```
msk-sashubina-gw-02# configure terminal
msk-sashubina-gw-02(config)# interface eth0
msk-sashubina-gw-02(config-if)# shutdown
msk-sashubina-gw-02(config-if)#
```

Рис. 47: gw-02

Отключила интерфейс eth0 на msk-sashubina-gw-04

```
msk-sashubina-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop                Via      Metric Tag Time
C(i) 2001:1::/64
      ::                self      1      0
R(n) 2001:2::/64
      fe80::ec8:7fff:fe62:0    eth1      2      0 02:39
R(n) 2001:3::/64
      fe80::e82:3aff:fec0:1    eth2      2      0 02:44
C(i) 2001:4::/64
      ::                self      1      0
C(i) 2001:10::/64
      ::                self      1      0
R(n) 2001:11::/64
      fe80::ec8:7fff:fe62:0    eth1      3      0 02:44
msk-sashubina-gw-01#
```

Рис. 48: gw-02

Отключила интерфейс eth0 на msk-sashubina-gw-04

```
msk-sashubina-gw-04# configure terminal
msk-sashubina-gw-04(config)# interface eth0
msk-sashubina-gw-04(config-if)# shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 49: gw-04

Отключила интерфейс eth0 на msk-sashubina-gw-04

```
msk-sashubina-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

   Network      Next Hop          Via      Metric Tag Time
C(i) 2001:1::/64      ::                self       1    0
R(n) 2001:2::/64      fe80::ec8:7fff:fe62:0    eth1       2    0  00:24
R(n) 2001:3::/64      fe80::e82:3aff:fec0:1    eth2      16    0  01:41
C(i) 2001:4::/64      ::                self       1    0
C(i) 2001:10::/64     ::                self       1    0
R(n) 2001:11::/64     fe80::ec8:7fff:fe62:0    eth1       3    0  02:35
msk-sashubina-gw-01#
```

Рис. 50: gw-04

Пропинговала PC2 с PC1 для проверки нового пути.

```
VPCS> ping 10.0.11.10  
  
*10.0.10.1 icmp_seq=1 ttl=64 time=1.249 ms (ICMP type:3, code:0, Destination net  
work unreachable)  
*10.0.10.1 icmp_seq=2 ttl=64 time=1.224 ms (ICMP type:3, code:0, Destination net  
work unreachable)  
*10.0.10.1 icmp_seq=3 ttl=64 time=1.152 ms (ICMP type:3, code:0, Destination net  
work unreachable)  
*10.0.10.1 icmp_seq=4 ttl=64 time=0.958 ms (ICMP type:3, code:0, Destination net  
work unreachable)  
10.0.11.10 icmp_seq=5 timeout  
  
VPCS> 
```

Рис. 51: PC2

Включила интерфейс:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# interface eth0
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 52: gw-04

Повторила пинг для проверки восстановления первоначального пути.

```
VPCS> ping 10.0.11.10  
  
10.0.11.10 icmp_seq=1 timeout  
10.0.11.10 icmp_seq=2 timeout  
10.0.11.10 icmp_seq=3 timeout  
10.0.11.10 icmp_seq=4 timeout  
10.0.11.10 icmp_seq=5 timeout  
  
VPCS> 
```

Рис. 53: ping

No.	Time	Source	Destination	Protocol	Length	Info
342	5601.473436	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x029, seq=2/512, ttl=64 (no response found)
343	5603.473618	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x029, seq=3/768, ttl=64 (no response found)
344	5605.474084	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x029, seq=4/1024, ttl=64 (no response found)
345	5607.475322	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x129, seq=5/1280, ttl=64 (no response found)
346	5609.539583	fe80::e4:1e:d9:00:00:00	fe80::f02:19	RIPng	186	Command Response, Version 1

▼ Frame 343: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Section number: 1

> Interface id: 0 (-)

Encapsulation type: Ethernet (1)

Arrival Time: Dec 15, 2025 18:28:13.978278000 RTZ 2 (same)

UTC Arrival Time: Dec 15, 2025 15:28:13.978278000 UTC

Epoch Arrival Time: 1765812493.978278000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 2.000222000 seconds]

[Time delta from previous displayed frame: 2.000222000 seconds]

[Time since reference or first frame: 5603.473658000 seconds]

Frame Number: 343

Frame Length: 98 bytes (784 bits)

Capture Length: 98 bytes (784 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:icmp:data]

[Coloring Rule Name: ICMP]

[Coloring Rule String: icmp || icmpv6]

▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:e4:1e:d9:00:00 (0c:e4:1e:d9:00:00)

Destination: 0c:e4:1e:d9:00:00 (0c:e4:1e:d9:00:00)

.....0 = LG bit: Globally unique address (factory default)

.....0 = IG bit: Individual address (unicast)

Source: Private_66:68:00 (00:50:79:66:68:00)

.....0 = LG bit: Globally unique address (factory default)

.....0 = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 8]

▼ Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10

0100 = Version: 4

.....0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Standard Input: -live capture in progress-

Пакеты: 433

Профиль: Default

Рис. 54: захват трафика

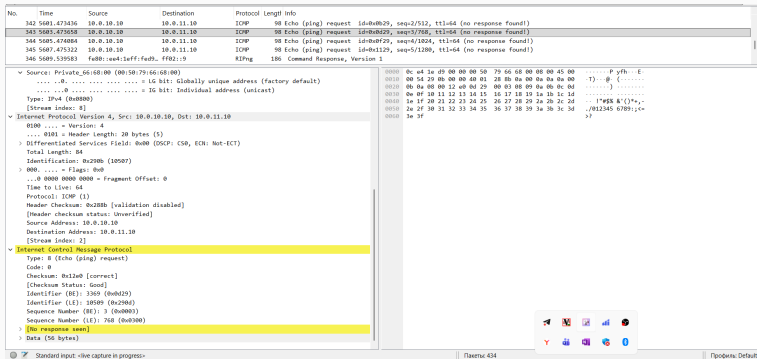


Рис. 55: захват трафика

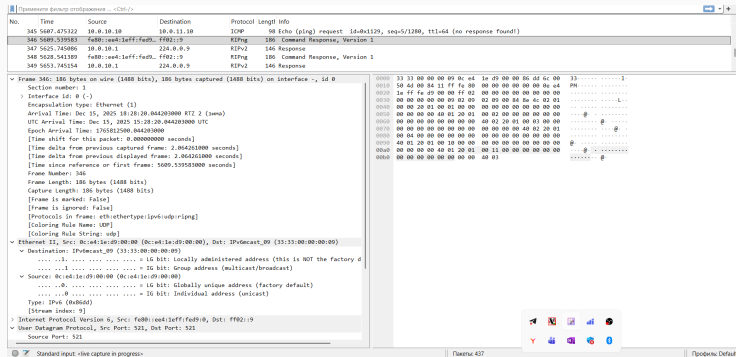


Рис. 56: захват трафика

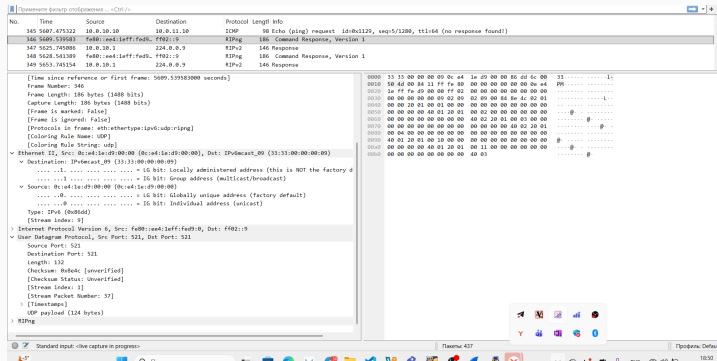


Рис. 57: захват трафика

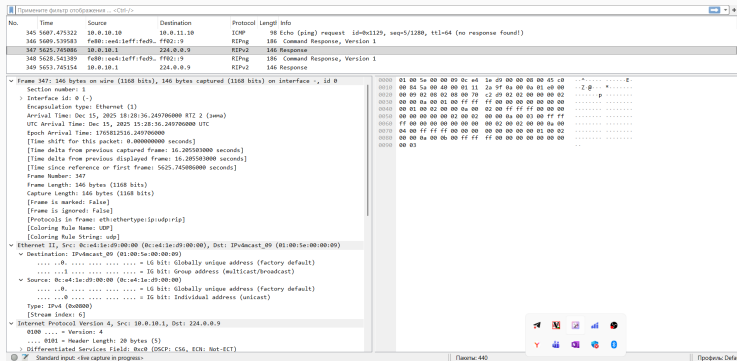


Рис. 58: захват трафика

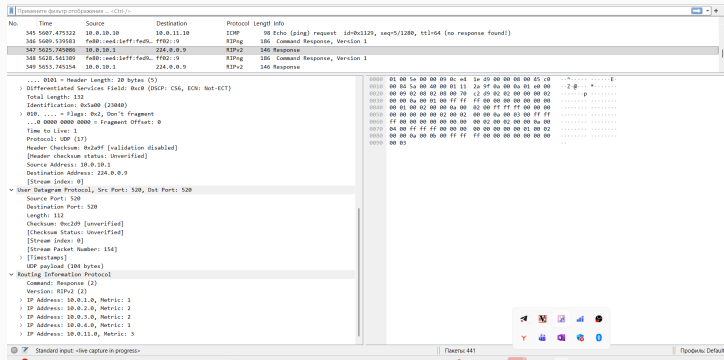


Рис. 59: захват трафика

Захват трафика

No.	Time	Source	Destination	Protocol	Length	Info
258	5548.801921	fa80::e11:94ff:fe4e...:ff02::9		RIPng	186	Command Response, Version 1
259	5549.155399	fa80::e11:94ff:fe4e...:ff02::9		RIPng	86	Command Response, Version 1
260	5576.172481	10.0.11.1	224.0.0.9	RIPv2	146	Response
261	5576.655990	10.0.11.1	224.0.0.9	RIPv2	86	Response

▼ frame 259: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface ..., id 0 Section number: 1 > Interface id: 0 (-) Encapsulation type: Ethernet (1) Arrival Time: Dec 15, 2025 18:27:19.775355000 RTZ 2 (sawa) UTC Arrival Time: Dec 15, 2025 15:27:19.775355000 UTC Epoch Arrival Time: 1765812439.775355000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 0.351478000 seconds] [Time delta from previous displayed frame: 0.351478000 seconds] [Time since reference or first frame: 5549.155399000 seconds] Frame Number: 259 Frame Length: 86 bytes (688 bits) Capture Length: 86 bytes (688 bits) [Frame is marked: False] [Frame is ignored: False] [Protocols in frame: eth:ethertype:ip:v6:udp:ripng] [Coloring Rule Name: UDP] [Coloring Rule String: udp] ▼ Ethernet II, Src: 0c:11:94:4e:00:00 (0c:11:94:4e:00:00), Dst: IPv6cast_09 (33:33:00:00:00:09) ▼ Destination: IPv6cast_09 (33:33:00:00:00:09) 1. = IG bit: Locally administered address (this is NOT the factory...) 1 = IG bit: Group address (multicast/broadcast) ▼ Source: 0c:11:94:4e:00:00 (0c:11:94:4e:00:00) 0. = IG bit: Globally unique address (factory default) 0 = IG bit: Individual address (unicast) Type: IPv6 (0x86dd) [Stream index: 10] ▼ Internet Protocol Version 6, Src: fa80::e11:94ff:fe4e:0, Dst: ff02::9 0110 = Version: 6 > 1100 0000 = Traffic Class: 0xc0 (DSCP: CS6, ECN: Not-ECT) 1100 0111 0001 1110 1111 = Flow Label: 0xc71ef Download Length: 32	0000 33 33 00 00 00 09 0c 11 94 4e 00 00 86 dd 6c 0c 33 M....1. 0010 71 ef 00 20 11 ff fe 00 00 00 00 00 00 00 00 00 4c 0020 94 ff fe 4e 00 00 ff 02 00 00 00 00 00 00 00 00 94 0030 00 00 00 00 00 00 02 09 02 09 00 20 fa 9a 02 01 0040 00 00 20 01 00 10 00 00 00 00 00 00 00 00 00 00 0050 00 00 00 00 40 03
---	---

Рис. 60: захват трафика

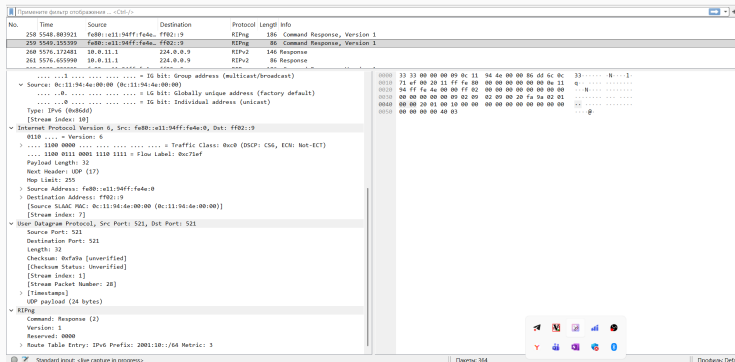


Рис. 61: захват трафика

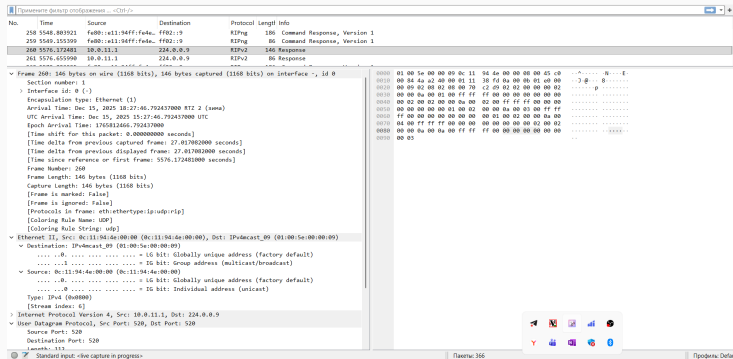


Рис. 62: захват трафика

Захват трафика

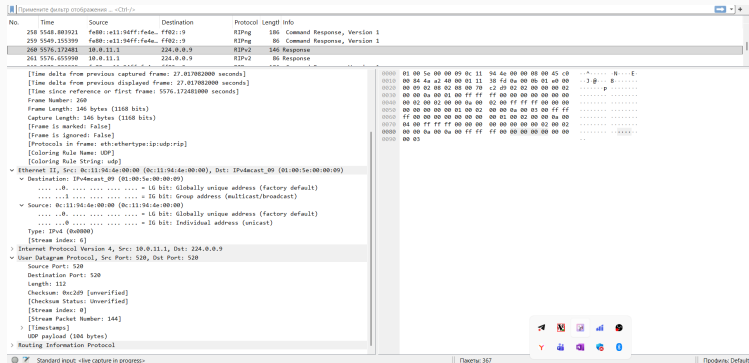


Рис. 63: захват трафика

4.1 Настройка OSPFv2 для IPv4

1. Я удалила конфигурацию RIP и настроила OSPFv2, объявив все сети в backbone-области (0.0.0.0).

```
msk-sashubina-gw-01# configure terminal
msk-sashubina-gw-01(config)# router ospf
msk-sashubina-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-sashubina-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-sashubina-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-sashubina-gw-01(config-router)# exit
msk-sashubina-gw-01(config)# exit
msk-sashubina-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-01#
```

Рис. 64: gw-01

Настройка для msk-sashubina-gw-02:

```
msk-sashubina-gw-02(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-02(config-if)# router ospf
msk-sashubina-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-sashubina-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-sashubina-gw-02(config-router)# exit
msk-sashubina-gw-02(config)# exit
msk-sashubina-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-02# █
```

Рис. 65: gw-02

Настройка для msk-sashubina-gw-03:

```
msk-sashubina-gw-03# configure terminal
msk-sashubina-gw-03(config)# router ospf
msk-sashubina-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-sashubina-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-sashubina-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-sashubina-gw-03(config-router)# exit
msk-sashubina-gw-03(config)# exit
msk-sashubina-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-03#
```

Рис. 66: gw-03

Настройка для msk-sashubina-gw-04:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# router ospf
msk-sashubina-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-sashubina-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-sashubina-gw-04(config-router)# exit
msk-sashubina-gw-04(config)# exit
msk-sashubina-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-04#
```

Рис. 67: gw-04

2. С PC1-sashubina я пропинговала PC2-sashubina и провела трассировку:

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=8.713 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=7.402 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=4.903 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=5.888 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=5.818 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    2.585 ms  0.615 ms  0.546 ms
 2  10.0.1.2     6.045 ms  7.263 ms  2.807 ms
 3  10.0.2.2     6.772 ms  4.276 ms  3.766 ms
 4  10.0.11.10   4.998 ms  4.992 ms  5.833 ms

VPCS> █
```

Рис. 68: PC1 ping

3. Я проверила состояние OSPF:

```
msk-sashubina-gw-01# show ip ospf neighbor
Neighbor ID      Pri State          Up Time      Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1          1 Full/Backup   4m11s        38.490s 10.0.1.2      et
h1:10.0.1.1      0          0          0
10.0.4.1          1 Full/Backup   1m24s        33.579s 10.0.4.1      et
h2:10.0.4.2      0          0          0

msk-sashubina-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
                        directly attached to eth1
N   10.0.2.0/24      [200] area: 0.0.0.0
                        via 10.0.1.2, eth1
N   10.0.3.0/24      [200] area: 0.0.0.0
                        via 10.0.4.1, eth2
N   10.0.4.0/24      [100] area: 0.0.0.0
                        directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
                        directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
                        via 10.0.1.2, eth1
                        via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-sashubina-gw-01#
```

Рис. 69: gw-01

4. Отключила интерфейс eth0 на msk-sashubina-gw-04:

```
msk-sashubina-gw-04# configure terminal
msk-sashubina-gw-04 (config)# interface eth0
msk-sashubina-gw-04 (config-if)# shutdown
msk-sashubina-gw-04 (config-if)#
```

Рис. 70: gw-04

5. Проверила таблицы OSPF на msk-sashubina-gw-01.

```
msk-sashubina-gw-01# show ip ospf neighbor
Neighbor ID      Pri State           Up Time           Dead Time Address      Interface
10.0.2.1         1 Full/Backup      9m49s             39.941s 10.0.1.2       eth1:10.0.1.1
10.0.4.1         1 Full/Backup      7m02s             35.022s 10.0.4.1       eth2:10.0.4.2
                  RXmtL RqstL DBsmL
                  0     0     0
                  0     0     0

msk-sashubina-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
      directly attached to eth1
N   10.0.2.0/24      [200] area: 0.0.0.0
      via 10.0.1.2, eth1
N   10.0.3.0/24      [300] area: 0.0.0.0
      via 10.0.1.2, eth1
N   10.0.4.0/24      [100] area: 0.0.0.0
      directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
      directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
      via 10.0.1.2, eth1

===== OSPF router routing table =====
===== OSPF external routing table =====

msk-sashubina-gw-01#
```

Рис. 71: gw-01

6. Пропинговала PC2 с PC1.

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=5.327 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=5.891 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=6.466 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.635 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=3.659 ms

VPCS> █
```

Рис. 72: PC1 ping

7. Включила интерфейс обратно:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# interface eth0
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 73: gw-04

8. Выполнила пинг и трассировку для проверки восстановления пути.

```
VPCS> ping 10.0.11.10  
  
84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=10.244 ms  
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=7.425 ms  
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=5.072 ms  
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=6.225 ms  
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.861 ms  
  
VPCS> █
```

Рис. 74: PC1 ping

9. Захват трафика

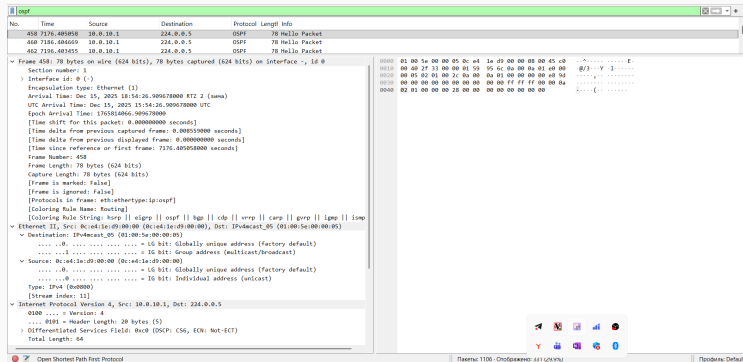


Рис. 75: захват трафика

9. Захват трафика

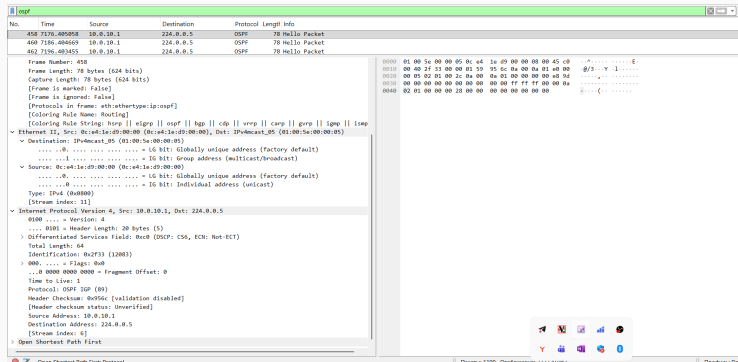


Рис. 76: захват трафика

9. Захват трафика

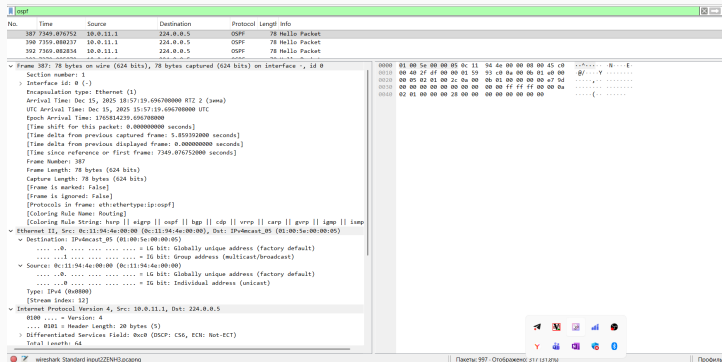


Рис. 77: захват трафика

9. Захват трафика

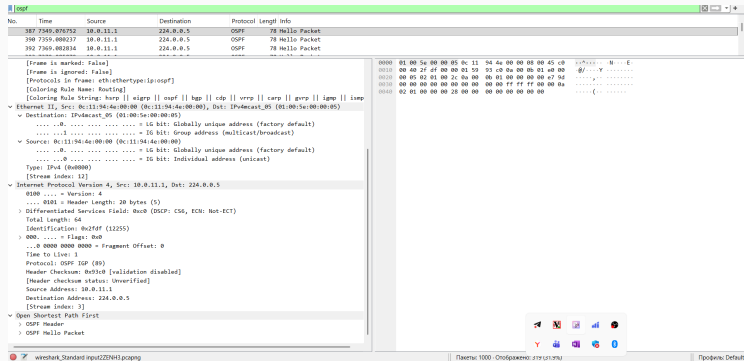


Рис. 78: захват трафика

10. Я настроила OSPFv3 для IPv6, задав Router ID и назначив интерфейсы в область 0.

```
msk-sashubina-gw-01# configure terminal
msk-sashubina-gw-01(config)# router ospf6
msk-sashubina-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-sashubina-gw-01(config-ospf6)# exit
msk-sashubina-gw-01(config)# interface eth0
msk-sashubina-gw-01(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-01(config-if)# exit
msk-sashubina-gw-01(config)# interface eth1
msk-sashubina-gw-01(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-01(config-if)# exit
msk-sashubina-gw-01(config)# interface eth2
msk-sashubina-gw-01(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-01(config-if)# exit
msk-sashubina-gw-01(config)# exit
msk-sashubina-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-01#
```

Рис. 79: gw-01

Настройка для msk-sashubina-gw-02:

```
msk-sashubina-gw-02(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-02(config-if)# router ospf6
msk-sashubina-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-sashubina-gw-02(config-ospf6)# exit
msk-sashubina-gw-02(config)# interface eth0
msk-sashubina-gw-02(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-02(config-if)# exit
msk-sashubina-gw-02(config)# interface eth1
msk-sashubina-gw-02(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-02(config-if)# exit
msk-sashubina-gw-02(config)# exit
msk-sashubina-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-02#
```

Рис. 80: gw-02

Настройка для msk-sashubina-gw-03:

```
msk-sashubina-gw-03# configure terminal
msk-sashubina-gw-03(config)# router ospf6
msk-sashubina-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-sashubina-gw-03(config-ospf6)# exit
msk-sashubina-gw-03(config)# interface eth0
msk-sashubina-gw-03(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# interface eth1
msk-sashubina-gw-03(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# interface eth2
msk-sashubina-gw-03(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-03(config-if)# exit
msk-sashubina-gw-03(config)# exit
msk-sashubina-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-03#
```

Рис. 81: gw-03

Настройка для msk-sashubina-gw-04:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# router ospf6
msk-sashubina-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-sashubina-gw-04(config-ospf6)# exit
msk-sashubina-gw-04(config)# interface eth0
msk-sashubina-gw-04(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-04(config-if)# exit
msk-sashubina-gw-04(config)# interface eth1
msk-sashubina-gw-04(config-if)# ipv6 ospf6 area 0
msk-sashubina-gw-04(config-if)# exit
msk-sashubina-gw-04(config)# exit
msk-sashubina-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-sashubina-gw-04#
```

Рис. 82: gw-04

11. Проверила связность IPv6:

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=7.041 ms
2001:11::a icmp6_seq=2 ttl=58 time=6.824 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.607 ms
2001:11::a icmp6_seq=4 ttl=58 time=6.860 ms
2001:11::a icmp6_seq=5 ttl=58 time=7.505 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1   3.201 ms  0.866 ms  1.169 ms
 2 2001:4::1    3.142 ms  2.324 ms  1.547 ms
 3 2001:3::1    5.553 ms  6.234 ms  4.177 ms
 4 2001:11::a   6.119 ms  4.152 ms  4.220 ms

VPCS> 
```

Рис. 83: ping

12. Проверила соседство и маршруты OSPFv3:

```
msk-sashubina-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri      DeadTime      State/IfState      Duration I/F[State]
2.2.2.2          1        00:00:35      Full/BDR          00:06:17 eth1[DR]
4.4.4.4          1        00:00:36      Full/BDR          00:01:09 eth2[DR]
msk-sashubina-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64      ::              eth1 00:01:20
*N IA 2001:2::/64      fe80::ec8:7fff:fe62:0 eth1 00:01:20
*N IA 2001:3::/64      fe80::e82:3aff:fec0:1 eth2 00:01:20
*N IA 2001:4::/64      ::              eth2 00:01:20
*N IA 2001:10::/64     ::              eth0 00:01:20
*N IA 2001:11::/64     fe80::ec8:7fff:fe62:0 eth1 00:01:20
                        fe80::e82:3aff:fec0:1 eth2
msk-sashubina-gw-01#
```

Рис. 84: gw-01

13. Отключила интерфейс eth0 на msk-sashubina-gw-04:

```
msk-sashubina-gw-04# configure terminal
msk-sashubina-gw-04(config)# interface eth0
msk-sashubina-gw-04(config-if)# shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 85: gw-04

14. Проверила таблицу маршрутизации OSPFv3 на msk-sashubina-gw-01. Маршрут мгновенно изменился.

```
msk-sashubina-gw-01# show ipv6 ospf6 route
*  
*N IA 2001:1::/64 :: eth1 00:00:32  
*N IA 2001:2::/64 fe80::ec8:7fff:fe62:0 eth1 00:00:32  
*N IA 2001:3::/64 fe80::ec8:7fff:fe62:0 eth1 00:00:32  
*N IA 2001:4::/64 :: eth2 00:00:32  
*N IA 2001:10::/64 :: eth0 00:00:32  
*N IA 2001:11::/64 fe80::ec8:7fff:fe62:0 eth1 00:00:32  
msk-sashubina-gw-01#
```

Рис. 86: gw-01

15. Пропинговала PC2 с PC1.

```
VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=5.688 ms
2001:11::a icmp6_seq=2 ttl=58 time=5.462 ms
2001:11::a icmp6_seq=3 ttl=58 time=4.228 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.251 ms
2001:11::a icmp6_seq=5 ttl=58 time=8.744 ms

VPCS> trace 2001:11::a
trace to 2001:11::a, 64 hops max
 1 2001:10::1 0.987 ms 0.445 ms 1.753 ms
 2 2001:1::2 5.294 ms 3.224 ms 1.997 ms
 3 2001:2::2 4.406 ms 3.554 ms 4.083 ms
 4 2001:11::a 6.203 ms 4.735 ms 5.372 ms

VPCS> █
```

Рис. 87: ping

16. Включила интерфейс:

```
msk-sashubina-gw-04(config-if)# configure terminal
% Unknown command: configure terminal
msk-sashubina-gw-04(config-if)# interface eth0
msk-sashubina-gw-04(config-if)# no shutdown
msk-sashubina-gw-04(config-if)#
```

Рис. 88: gw-04

17. Убедилась в восстановлении первоначального пути через пинг и трассировку.

```
VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=7.527 ms
2001:11::a icmp6_seq=2 ttl=58 time=4.091 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.494 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.870 ms
2001:11::a icmp6_seq=5 ttl=58 time=6.689 ms

VPCS> trace 2001:11::a
trace to 2001:11::a, 64 hops max
 1 2001:10::1  0.875 ms  0.602 ms  0.467 ms
 2 2001:4::1   5.344 ms  5.171 ms  2.126 ms
 3 2001:3::1   9.907 ms  7.268 ms  5.064 ms
 4 2001:11::a  8.327 ms  9.729 ms  6.893 ms

VPCS> 
```

Рис. 89: ping

18. Захват трафика

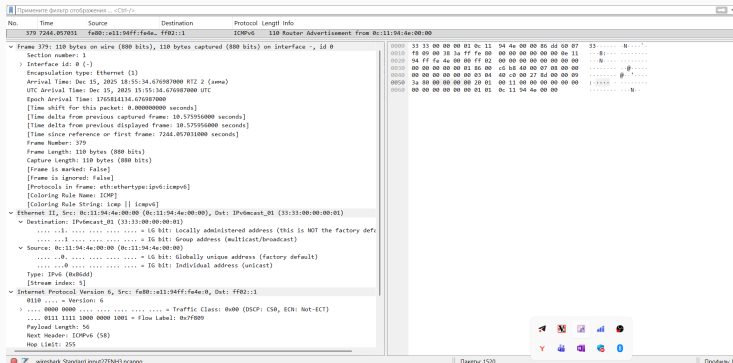


Рис. 90: анализ трафика

18. Захват трафика

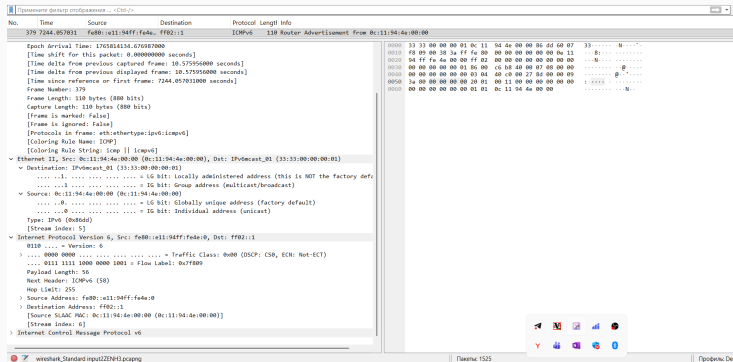


Рис. 91: анализ трафика

18. Захват трафика

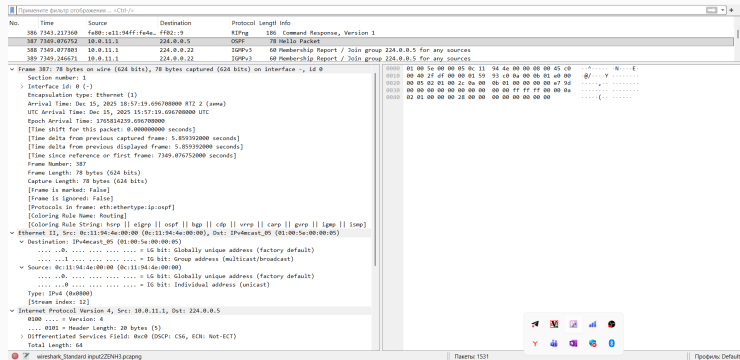


Рис. 92: анализ трафика

18. Захват трафика

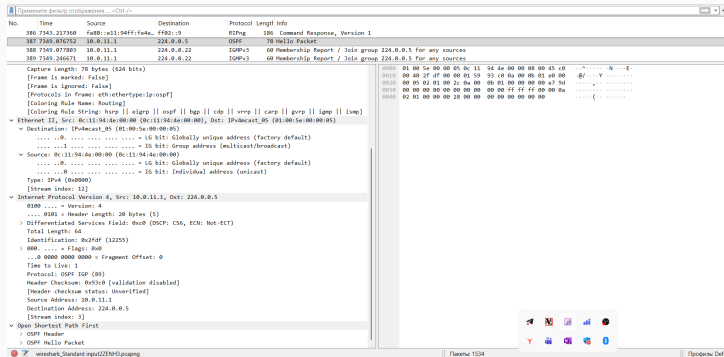


Рис. 93: анализ трафика

18. Захват трафика

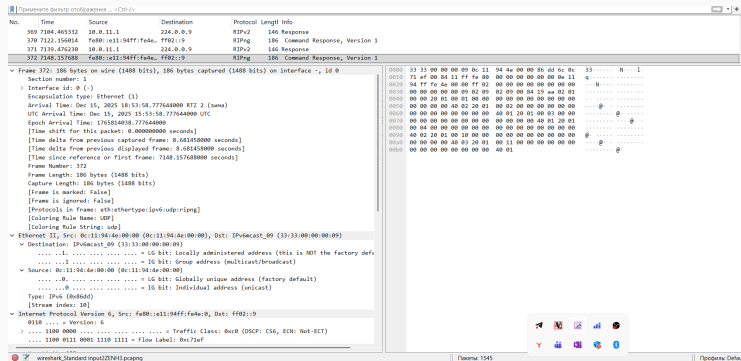


Рис. 94: анализ трафика

18. Захват трафика

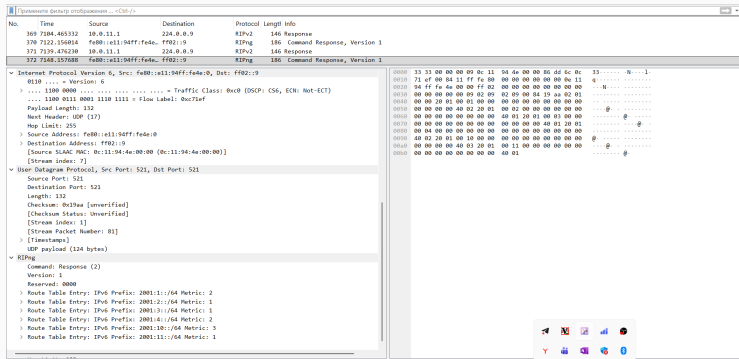


Рис. 95: анализ трафика

18. Захват трафика

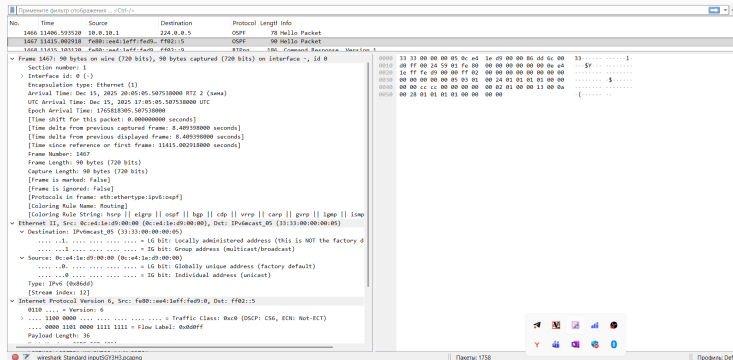


Рис. 96: анализ трафика

18. Захват трафика

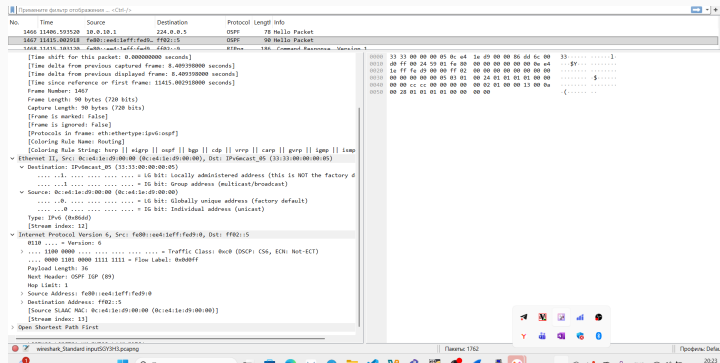


Рис. 97: анализ трафика

18. Захват трафика

Примените фильтр отображения: «См-?»

No.	Time	Source	Destination	Protocol	Length	Info
1466	11406.593520	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1467	11415.002910	fe80::e4:1eff:fe9::ff02::5		OSPF	90	Hello Packet
1468	11415.103120	fe80::e4:1eff:fe9::ff02::9		RIPng	186	Command Response, Version 1
1469	11416.596400	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1470	11425.006310	fe80::e4:1eff:fe9::ff02::5		OSPF	90	Hello Packet

▼ Frame 1469: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface -, id 0

Section number: 1

Interface id: 0 (-)

Encapsulation type: Ethernet (1)

Arrival Time: Dec 15, 2025 20:05:07.101118000 RTZ 2 (asw)

UTC Arrival Time: Dec 15, 2025 17:05:07.101118000 UTC

Epoch Arrival Time: 1765818307.101118000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 1.693378000 seconds]

[Time delta from previous displayed frame: 1.493378000 seconds]

[Time since reference or first frame: 11416.596400000 seconds]

Frame Number: 1469

Frame Length: 78 bytes (624 bits)

Capture Length: 78 bytes (624 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: ethertype:ip:ospf]

[Coloring Rule Name: Routing]

[Coloring Rule String: hsrp || eigrp || ospf || bgp || cdp || vrrp || carp || gvrp || igmp || icmp

▼ Ethernet II, Src: 0c:e4:1e:d9:00:00 (0c:e4:1e:d9:00:00), Dst: Ipv6cast_05 (01:00:5e:00:00:05)

▼ Destination: Ipv6cast_05 (01:00:5e:00:00:05)

..... = 16 bit: Globally unique address (factory default)

.....3..... = 16 bit: Group address (multicast/broadcast)

▼ Source: 0c:e4:1e:d9:00:00 (0c:e4:1e:d9:00:00)

..... = 16 bit: Globally unique address (factory default)

.....0..... = 16 bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 11]

▼ Internet Protocol Version 4, Src: 10.0.10.1, Dst: 224.0.0.5

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

0000 01 00 5e 00 00 05 0c e4 1e d9 00 00 00 45 c0E-

0010 00 60 34 02 00 00 01 50 00 1d 0a 00 0a 01 a0 00 04.....Y.....

0020 00 05 02 01 00 2c 0a 00 0a 01 00 00 00 44 0c{.....

0030 00 00 00 00 00 00 00 00 00 00 ff ff ff 00 0a{.....

0040 02 01 00 00 00 28 0a 00 0a 01 00 00 00 00{.....

Рис. 98: анализ трафика

18. Захват трафика

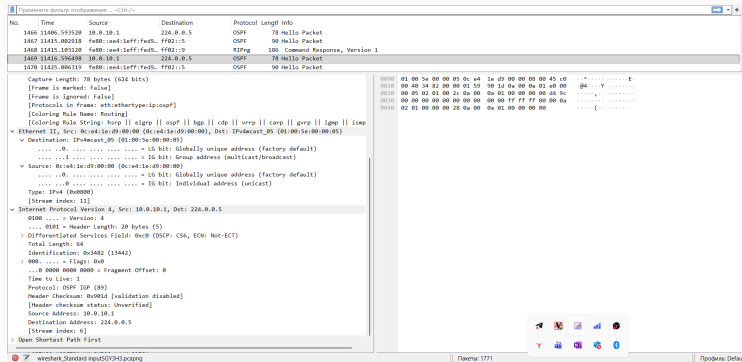


Рис. 99: анализ трафика

18. Захват трафика

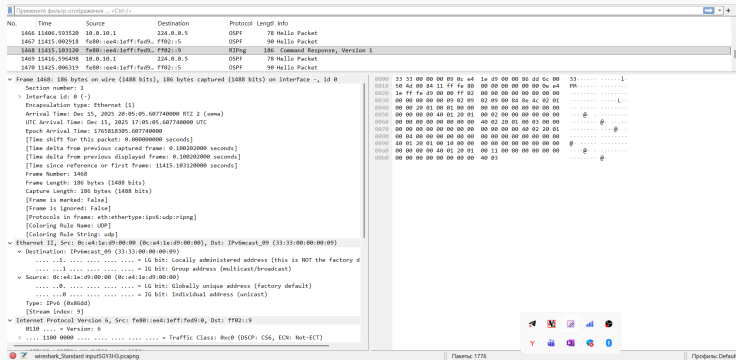


Рис. 100: анализ трафика

18. Захват трафика

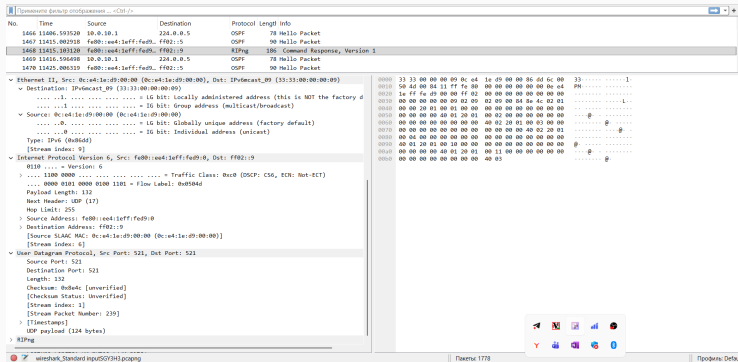


Рис. 101: анализ трафика

5.1 Создание проекта и базовой настройки

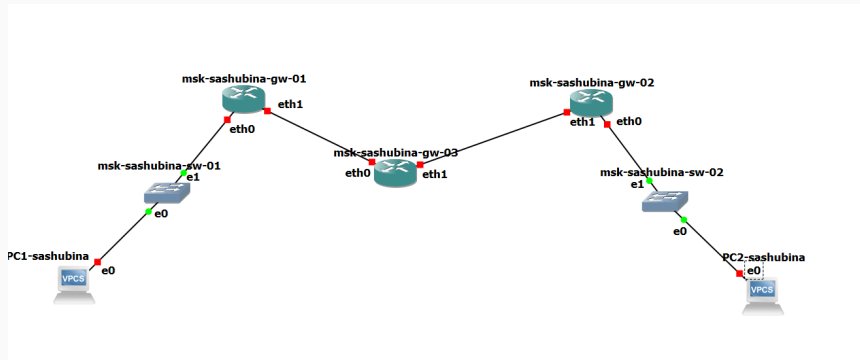


Рис. 102: создание проекта

4. Я подключила анализатор трафика на соединении между первым и третьим маршрутизаторами.

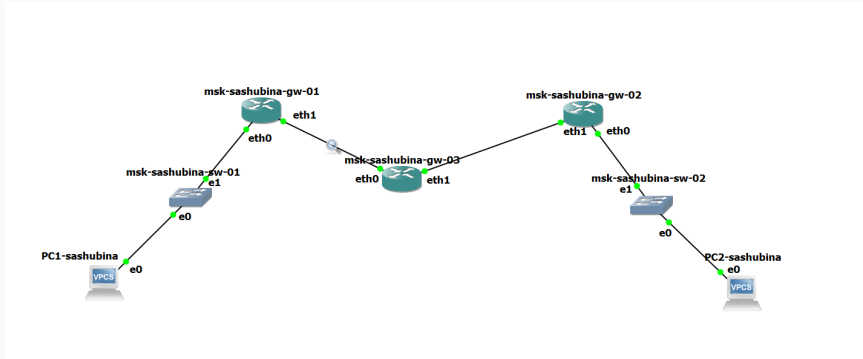


Рис. 103: анализ трафика

5. Присвоила адреса оконечным устройствам:

```
VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 20072
RHOST:PORT           : 127.0.0.1:20073
MTU                 : 1500

VPCS>
```

Рис. 104: PC1


```
VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 20074
RHOST:PORT          : 127.0.0.1:20075
MTU                 : 1500

VPCS> █
```

Рис. 105: PC2

7. Изменила имена устройств на маршрутизаторах:

```
vyos@msk-sashubina-gw-0x:~$ configure
[edit]
vyos@msk-sashubina-gw-0x# set system host-name msk-sashubina-gw-01
[edit]
vyos@msk-sashubina-gw-0x# compare
[edit system]
>host-name msk-sashubina-gw-01
[edit]
vyos@msk-sashubina-gw-0x# commit
[edit]
vyos@msk-sashubina-gw-0x# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-0x# exit
exit
vyos@msk-sashubina-gw-0x:~$ reboot
Are you sure you want to reboot this system? [y/N] █
```

Рис. 106: gw-01

7. Изменила имена устройств на маршрутизаторах:

```
vyos@msk-sashubina-gw-0x:~$ configure
[edit]
vyos@msk-sashubina-gw-0x# set system host-name msk-sashubina-gw-02
[edit]
vyos@msk-sashubina-gw-0x# compare
[edit system]
>host-name msk-sashubina-gw-02
[edit]
vyos@msk-sashubina-gw-0x# commit
[edit]
vyos@msk-sashubina-gw-0x# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-0x# exit
exit
vyos@msk-sashubina-gw-0x:~$ reboot
Are you sure you want to reboot this system? [y/N] █
```

Рис. 107: gw-02

7. Изменила имена устройств на маршрутизаторах:

```
vyos@msk-sashubina-gw-02:~$ configure
[edit]
vyos@msk-sashubina-gw-02# set system host-name msk-sashubina-gw-03
[edit]
vyos@msk-sashubina-gw-02# compare
[edit system]
>host-name msk-sashubina-gw-03
[edit]
vyos@msk-sashubina-gw-02# commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02# exit
exit
vyos@msk-sashubina-gw-02:~$ reboot
Are you sure you want to reboot this system? [y/N] █
```

Рис. 108: gw-03

5.2 Настройка адресов и проверка связи

```
vyos@msk-sashubina-gw-01:~$ configure
(edit)
vyos@msk-sashubina-gw-01# set interfaces ethernet eth0 address 1000::1/64

Configuration path: [interfaces ethernet eth0 address 1000::1/64] already exists

(edit)
vyos@msk-sashubina-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8

Configuration path: [interfaces ethernet eth1 address 10.0.0.1/8] already exists

(edit)
/64s@msk-sashubina-gw-01# set service router-advert interface eth0 prefix 1000::

Configuration path: [service router-advert interface eth0 prefix 1000::/64] already exists

(edit)
vyos@msk-sashubina-gw-01# commit
No configuration changes to commit
(edit)
vyos@msk-sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
(edit)
vyos@msk-sashubina-gw-01#
```

Рис. 109: gw-01

Для msk-sashubina-gw-02:

```
vyos@msk-sashubina-gw-02:~$ configure
[edit]
vyos@msk-sashubina-gw-02# set interfaces ethernet eth0 address 1002::1/64

Configuration path: [interfaces ethernet eth0 address 1002::1/64] already exists

[edit]
vyos@msk-sashubina-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8

Configuration path: [interfaces ethernet eth1 address 20.0.0.2/8] already exists

[edit]
/64s@msk-sashubina-gw-02# set service router-advert interface eth0 prefix 1002::

Configuration path: [service router-advert interface eth0 prefix 1002::/64] already exists

[edit]
vyos@msk-sashubina-gw-02# commit
No configuration changes to commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02# █
```

Рис. 110: gw-02

Для msk-sashubina-gw-03:

```
vyos@msk-sashubina-gw-03:~$ configure
[edit]
vyos@msk-sashubina-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8

Configuration path: [interfaces ethernet eth0 address 10.0.0.2/8] already exists

[edit]
vyos@msk-sashubina-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8

Configuration path: [interfaces ethernet eth1 address 20.0.0.1/8] already exists

[edit]
vyos@msk-sashubina-gw-03# commit
No configuration changes to commit
[edit]
vyos@msk-sashubina-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-03# █
```

Рис. 111: gw-03

9. Проверила, что на PC появились адреса ближайших маршрутизаторов:

```
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                 :
ROUTER LINK-LAYER   : 0c:6a:85:74:00:00
MAC                 : 00:50:79:66:68:00
LPORT               : 20072
RHOST:PORT           : 127.0.0.1:20073
MTU                  : 1500

VPCS> █
```

Рис. 112: PC1

9. Проверила, что на PC появились адреса ближайших маршрутизаторов:

```
VPCS> show ipv6  
  
NAME                : VPCS[1]  
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64  
GLOBAL SCOPE        : 1002::a/64  
DNS                  :  
ROUTER LINK-LAYER   : 0c:f6:01:a2:00:00  
MAC                  : 00:50:79:66:68:01  
LPORT                : 20074  
RHOST:PORT           : 127.0.0.1:20075  
MTU                  : 1500  
  
VPCS> █
```

Рис. 113: PC2

10. Проверила маршруты с маршрутизатора R1:

```
vyos@uk-sashubina-qw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=28.5 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=2.01 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=12.1 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=19.6 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=8.47 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=14.0 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=15.8 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=15.4 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=12.7 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=7.64 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=3.31 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=25.3 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=8.52 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=7.05 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=5.75 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=3.43 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=6.30 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=9.87 ms
^C
[1]* Stopped ping 10.0.0.2
[edit]
vyos@uk-sashubina-qw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data:
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=18.9 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=9.92 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=1.55 ms
64 bytes from 20.0.0.1: icmp_seq=4 ttl=64 time=15.6 ms
64 bytes from 20.0.0.1: icmp_seq=5 ttl=64 time=15.3 ms
64 bytes from 20.0.0.1: icmp_seq=6 ttl=64 time=14.3 ms
64 bytes from 20.0.0.1: icmp_seq=7 ttl=64 time=14.7 ms
64 bytes from 20.0.0.1: icmp_seq=8 ttl=64 time=15.0 ms
64 bytes from 20.0.0.1: icmp_seq=9 ttl=64 time=1.72 ms
^C
[1]* Stopped ping 20.0.0.1
[edit]
vyos@uk-sashubina-qw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data:
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=16.8 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=11.3 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=9.35 ms
64 bytes from 20.0.0.2: icmp_seq=4 ttl=63 time=3.47 ms
^C
[3]* Stopped ping 20.0.0.2
[edit]
vyos@uk-sashubina-qw-01#
```

Рис. 114: ping

5.3 Настройка RIP маршрутизации для IPv4

```
vyos@msk-sashubina-gw-01# set protocols rip network 10.0.0.0/8

  Configuration path: [protocols rip network 10.0.0.0/8] already exists

[edit]
vyos@msk-sashubina-gw-01# commit
No configuration changes to commit
[edit]
vyos@msk-sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-01#
```

Рис. 115: gw-01

Для msk-sashubina-gw-02:

```
vyos@msk-sashubina-gw-02# set protocols rip network 20.0.0.0/8

  Configuration path: [protocols rip network 20.0.0.0/8] already exists

[edit]
vyos@msk-sashubina-gw-02# commit
No configuration changes to commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02#
```

Рис. 116: gw-02

Для msk-sashubina-gw-03:

```
vyos@msk-sashubina-gw-03# set protocols rip network 10.0.0.0/8

Configuration path: [protocols rip network 10.0.0.0/8] already exists

[edit]
vyos@msk-sashubina-gw-03# set protocols rip network 20.0.0.0/8

Configuration path: [protocols rip network 20.0.0.0/8] already exists

[edit]
vyos@msk-sashubina-gw-03# commit
No configuration changes to commit
[edit]
vyos@msk-sashubina-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-03#
```

Рис. 117: gw-03

12. Проверила маршруты после настройки RIP:

```
vyos@msk-sashubina-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=14.3 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=5.15 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=7.48 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=3.68 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=15.1 ms
^C
[4]+  Stopped                  ping 10.0.0.2
[edit]
vyos@msk-sashubina-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=19.2 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=14.5 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=4.00 ms
64 bytes from 20.0.0.1: icmp_seq=4 ttl=64 time=15.9 ms
64 bytes from 20.0.0.1: icmp_seq=5 ttl=64 time=8.91 ms
64 bytes from 20.0.0.1: icmp_seq=6 ttl=64 time=2.12 ms
64 bytes from 20.0.0.1: icmp_seq=7 ttl=64 time=13.5 ms
^C
[5]+  Stopped                  ping 20.0.0.1
[edit]
vyos@msk-sashubina-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=33.6 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=9.46 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=3.85 ms
^C
[6]+  Stopped                  ping 20.0.0.2
[edit]
vyos@msk-sashubina-gw-01#
```

Рис. 118: ping

5.4 Создание IPv6 туннеля через IPv4 сеть

```
vyos@msk-sashubina-gw-01# configure
Invalid command: [configure]
[edit]
vyos@msk-sashubina-gw-01# set interfaces tunnel tun0 encapsulation sit
Configuration path: [interfaces tunnel tun0 encapsulation sit] already exists
[edit]
vyos@msk-sashubina-gw-01# set interfaces tunnel tun0
Configuration path: [interfaces tunnel tun0] already exists
[edit]
vyos@msk-sashubina-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
Configuration path: [interfaces tunnel tun0 source-address 10.0.0.1] already exists
[edit]
vyos@msk-sashubina-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
Configuration path: [interfaces tunnel tun0 remote 20.0.0.2] already exists
[edit]
vyos@msk-sashubina-gw-01# set interfaces tunnel tun0 address 1001::1/64
Configuration path: [interfaces tunnel tun0 address 1001::1/64] already exists
[edit]
vyos@msk-sashubina-gw-01# commit
No configuration changes to commit
[edit]
vyos@msk-sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-01#
```

Рис. 119: gw-01

Для msk-sashubina-gw-02:

```
vyos@msk-sashubina-gw-02# configure

Invalid command: [configure]

[edit]
vyos@msk-sashubina-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-sashubina-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-sashubina-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-sashubina-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-sashubina-gw-02# commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02#
```

Рис. 120: gw-02

14. Настроила статическую маршрутизацию IPv6:

```
vyos@msk-sashubina-gw-01# configure

Invalid command: [configure]

[edit]
vyos@msk-sashubina-gw-01# set protocols static route6 1002::0/64 next-hop 1001::
2
[edit]
vyos@msk-sashubina-gw-01# commit
[edit]
vyos@msk-sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-01#
```

Рис. 121: gw-01

Для msk-sashubina-gw-02:

```
vyos@msk-sashubina-gw-02# configure

  Invalid command: [configure]

[edit]
vyos@msk-sashubina-gw-02# set protocols static route6 1000::0/64 next-hop 1001::
1
[edit]
vyos@msk-sashubina-gw-02# commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02#
```

Рис. 122: gw-02

15. Проверила доступность оконечных устройств:

```
VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=29.976 ms
1002::a icmp6_seq=2 ttl=60 time=14.870 ms
1002::a icmp6_seq=3 ttl=60 time=6.793 ms
1002::a icmp6_seq=4 ttl=60 time=10.410 ms
1002::a icmp6_seq=5 ttl=60 time=11.745 ms

VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1    2.969 ms  2.352 ms  0.387 ms
 2 1001::2    8.507 ms  5.627 ms  7.429 ms
 3 1002::a    4.440 ms  1.875 ms  3.539 ms

VPCS> █
```

Рис. 123: PC1 ping

```
VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=8.397 ms
1000::a icmp6_seq=2 ttl=60 time=6.824 ms
1000::a icmp6_seq=3 ttl=60 time=10.065 ms
1000::a icmp6_seq=4 ttl=60 time=8.257 ms
1000::a icmp6_seq=5 ttl=60 time=4.609 ms

VPCS> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1    7.957 ms  0.670 ms  9.385 ms
 2 1001::1    12.552 ms 2.209 ms  5.703 ms
 3 1000::a    6.148 ms  2.469 ms  3.755 ms

VPCS> █
```

Рис. 124: PC2 ping

16. Анализ трафика

21	341.018465	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6ca67073
22	349.670945	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover - Transaction ID 0x6ca67073
23	474.998493	0c:00:7a:e3:00:01	Broadcast	ARP	42 Who has 10.0.0.2? Tell 10.0.0.1
24	475.001722	0c:0d:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 10.0.0.2 is at 0c:0d:79:dd:00:00
25	475.003021	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=1/256, ttl=64 (reply in 26)
26	475.004301	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=1/256, ttl=64 (request in 25)
27	475.999415	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=2/512, ttl=64 (reply in 28)
28	476.000161	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=2/512, ttl=64 (request in 27)
29	477.004836	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=3/768, ttl=64 (reply in 30)
30	477.006481	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=3/768, ttl=64 (request in 29)
31	478.006613	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=4/1024, ttl=64 (reply in 32)
32	478.007371	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=4/1024, ttl=64 (request in 31)
33	479.009342	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=5/1280, ttl=64 (reply in 34)
34	479.010056	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=5/1280, ttl=64 (request in 33)
35	480.011128	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=6/1536, ttl=64 (reply in 36)
36	480.012304	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=6/1536, ttl=64 (request in 35)
37	480.381333	0c:0d:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 Who has 10.0.0.1? Tell 10.0.0.2
38	480.383395	0c:00:7a:e3:00:01	0c:0d:79:dd:00:00	ARP	42 10.0.0.1 is at 0c:00:7a:e3:00:01
39	481.012519	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=7/1792, ttl=64 (reply in 40)
40	481.013487	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=7/1792, ttl=64 (request in 39)
41	482.014540	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=8/2048, ttl=64 (reply in 42)
42	482.015289	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=8/2048, ttl=64 (request in 41)
43	483.016211	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=9/2304, ttl=64 (reply in 44)
44	483.018013	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=9/2304, ttl=64 (request in 43)
45	484.016432	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0d71, seq=10/2560, ttl=64 (reply in 46)
46	484.017464	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0d71, seq=10/2560, ttl=64 (request in 45)

Рис. 125: анализ трафика

16. Анализ трафика

53	855.040525	10.0.0.2	224.0.0.9	RIPv2	66 Response
54	877.042110	10.0.0.2	224.0.0.9	RIPv2	66 Response
55	908.045669	10.0.0.2	224.0.0.9	RIPv2	66 Response
56	935.051920	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0e39, seq=1/256, ttl=64 (reply in 57)
57	935.053068	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e39, seq=1/256, ttl=64 (request in 56)
58	936.054174	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0e39, seq=2/512, ttl=64 (reply in 59)
59	936.055040	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e39, seq=2/512, ttl=64 (request in 58)
60	937.055995	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0e39, seq=3/768, ttl=64 (reply in 61)
61	937.057346	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e39, seq=3/768, ttl=64 (request in 60)
62	938.057904	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0e39, seq=4/1024, ttl=64 (reply in 63)
63	938.060024	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e39, seq=4/1024, ttl=64 (request in 62)
64	939.059497	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0e39, seq=5/1280, ttl=64 (reply in 65)
65	939.060251	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e39, seq=5/1280, ttl=64 (request in 64)
66	940.061767	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request id=0x0e39, seq=6/1536, ttl=64 (reply in 67)
67	940.062557	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e39, seq=6/1536, ttl=64 (request in 66)
68	941.032933	0c:00:7a:e3:00:01	0c:0d:79:dd:00:00	ARP	42 Who has 10.0.0.2? Tell 10.0.0.1
69	941.033616	0c:0d:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 10.0.0.2 is at 0c:0d:79:dd:00:00
70	941.051843	10.0.0.2	224.0.0.9	RIPv2	66 Response
71	941.180097	0c:0d:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 Who has 10.0.0.1? Tell 10.0.0.2
72	941.181022	0c:00:7a:e3:00:01	0c:0d:79:dd:00:00	ARP	42 10.0.0.1 is at 0c:00:7a:e3:00:01
73	946.591324	10.0.0.1	20.0.0.1	ICMP	98 Echo (ping) request id=0x0e3a, seq=1/256, ttl=64 (reply in 74)
74	946.592055	20.0.0.1	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3a, seq=1/256, ttl=64 (request in 73)
75	947.593695	10.0.0.1	20.0.0.1	ICMP	98 Echo (ping) request id=0x0e3a, seq=2/512, ttl=64 (reply in 76)
76	947.595214	20.0.0.1	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3a, seq=2/512, ttl=64 (request in 75)
77	948.595007	10.0.0.1	20.0.0.1	ICMP	98 Echo (ping) request id=0x0e3a, seq=3/768, ttl=64 (reply in 78)
78	948.595558	20.0.0.1	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3a, seq=3/768, ttl=64 (request in 77)
79	949.597219	10.0.0.1	20.0.0.1	ICMP	98 Echo (ping) request id=0x0e3a, seq=4/1024, ttl=64 (reply in 80)
80	949.598455	20.0.0.1	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3a, seq=4/1024, ttl=64 (request in 79)
81	950.600395	10.0.0.1	20.0.0.1	ICMP	98 Echo (ping) request id=0x0e3a, seq=5/1280, ttl=64 (reply in 82)
82	950.601107	20.0.0.1	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3a, seq=5/1280, ttl=64 (request in 81)
83	954.297539	10.0.0.1	20.0.0.2	ICMP	98 Echo (ping) request id=0x0e3b, seq=1/256, ttl=64 (reply in 84)
84	954.304665	20.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3b, seq=1/256, ttl=64 (request in 83)
85	955.299636	10.0.0.1	20.0.0.2	ICMP	98 Echo (ping) request id=0x0e3b, seq=2/512, ttl=64 (reply in 86)
86	955.303229	20.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply id=0x0e3b, seq=2/512, ttl=64 (request in 85)
87	970.052332	10.0.0.2	224.0.0.9	RIPv2	66 Response

Рис. 126: анализ трафика

16. Анализ трафика

102	1366.514435	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0x763e, seq=1, hop limit=61 (request in 101)
103	1367.521626	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0x763e, seq=2, hop limit=63 (reply in 104)
104	1367.524625	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0x763e, seq=2, hop limit=61 (request in 103)
105	1368.528948	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0x763e, seq=3, hop limit=63 (reply in 106)
106	1368.531115	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0x763e, seq=3, hop limit=61 (request in 105)
107	1369.533441	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0x763e, seq=4, hop limit=63 (reply in 108)
108	1369.536928	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0x763e, seq=4, hop limit=61 (request in 107)
109	1370.539558	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0x763e, seq=5, hop limit=63 (reply in 110)
110	1370.541556	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0x763e, seq=5, hop limit=61 (request in 109)
111	1371.625911	0c:00:7a:e3:00:01	0c:00:79:dd:00:00	ARP	42 Who has 10.0.0.2? Tell 10.0.0.1
112	1371.626679	0c:00:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 10.0.0.2 is at 0c:00:79:dd:00:00
113	1371.772795	0c:00:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 Who has 10.0.0.1? Tell 10.0.0.2
114	1371.774151	0c:00:7a:e3:00:01	0c:00:79:dd:00:00	ARP	42 10.0.0.1 is at 0c:00:7a:e3:00:01
115	1377.098816	1000::a	1002::a	UDP	146 11057 → 11058 len=64
116	1377.103413	1001::2	1000::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)
117	1377.105707	1000::a	1002::a	UDP	146 11057 → 11058 len=64
118	1377.107149	1001::2	1000::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)
119	1377.110204	1000::a	1002::a	UDP	146 11057 → 11058 len=64
120	1377.111625	1001::2	1000::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)
121	1377.113172	1000::a	1002::a	UDP	146 11057 → 11058 len=64
122	1377.116080	1002::a	1000::a	ICMPv6	194 Destination Unreachable (Port unreachable)[Malformed Packet]
123	1377.117952	1000::a	1002::a	UDP	146 11057 → 11058 len=64
124	1377.119488	1002::a	1000::a	ICMPv6	194 Destination Unreachable (Port unreachable)[Malformed Packet]
125	1377.122643	1000::a	1002::a	UDP	146 11057 → 11058 len=64
126	1377.127124	1002::a	1000::a	ICMPv6	194 Destination Unreachable (Port unreachable)[Malformed Packet]
127	1394.098944	10.0.0.2	224.0.0.9	RIPv2	66 Response
128	1415.411878	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=1, hop limit=63 (reply in 129)
129	1415.414735	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=1, hop limit=61 (request in 128)
130	1416.419902	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=2, hop limit=63 (reply in 131)
131	1416.422396	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=2, hop limit=61 (request in 130)
132	1417.429487	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=3, hop limit=63 (reply in 133)
133	1417.432984	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=3, hop limit=61 (request in 132)
134	1418.438912	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=4, hop limit=63 (reply in 135)
135	1418.441378	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=4, hop limit=61 (request in 134)
136	1419.447039	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=5, hop limit=63 (reply in 137)
137	1419.448449	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=5, hop limit=61 (request in 136)

Рис. 127: анализ трафика

16. Анализ трафика

127	1394.998944	10.0.0.2	224.0.0.9	RIPv2	66 Response
128	1415.411878	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=1, hop limit=63 (reply in 129)
129	1415.414735	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=1, hop limit=61 (request in 128)
130	1416.419902	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=2, hop limit=63 (reply in 131)
131	1416.422396	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=2, hop limit=61 (request in 130)
132	1417.429487	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=3, hop limit=63 (reply in 133)
133	1417.432984	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=3, hop limit=61 (request in 132)
134	1418.438912	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=4, hop limit=63 (reply in 135)
135	1418.441378	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=4, hop limit=61 (request in 134)
136	1419.447039	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xa73e, seq=5, hop limit=63 (reply in 137)
137	1419.448449	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xa73e, seq=5, hop limit=61 (request in 136)
138	1420.413263	0c:0d:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 Who has 10.0.0.1? Tell 10.0.0.2
139	1420.414122	0c:00:7a:e3:00:01	0c:0d:79:dd:00:00	ARP	42 10.0.0.1 is at 0c:00:7a:e3:00:01
140	1420.779985	0c:00:7a:e3:00:01	0c:0d:79:dd:00:00	ARP	42 Who has 10.0.0.2? Tell 10.0.0.1
141	1420.781074	0c:0d:79:dd:00:00	0c:00:7a:e3:00:01	ARP	42 10.0.0.2 is at 0c:0d:79:dd:00:00
142	1424.325063	1002::a	1000::a	UDOP	146 11057 → 11058 Len=64
143	1424.326467	1001::1	1002::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)
144	1424.331585	1002::a	1000::a	UDOP	146 11057 → 11058 Len=64
145	1424.332196	1001::1	1002::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)
146	1424.335975	1002::a	1000::a	UDOP	146 11057 → 11058 Len=64
147	1424.336633	1001::1	1002::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)
148	1424.339503	1002::a	1000::a	UDOP	146 11057 → 11058 Len=64
149	1424.341744	1000::a	1002::a	ICMPv6	194 Destination Unreachable (Port unreachable)[Malformed Packet]
150	1424.345972	1002::a	1000::a	UDOP	146 11057 → 11058 Len=64
151	1424.346604	1000::a	1002::a	ICMPv6	194 Destination Unreachable (Port unreachable)[Malformed Packet]
152	1424.348713	1002::a	1000::a	UDOP	146 11057 → 11058 Len=64
153	1424.349359	1000::a	1002::a	ICMPv6	194 Destination Unreachable (Port unreachable)[Malformed Packet]
154	1429.103836	10.0.0.2	224.0.0.9	RIPv2	66 Response
155	1464.108250	10.0.0.2	224.0.0.9	RIPv2	66 Response

Рис. 128: анализ трафика

8.3.3 Задание для самостоятельного выполнения

1. Задокументировала моделируемую сеть:

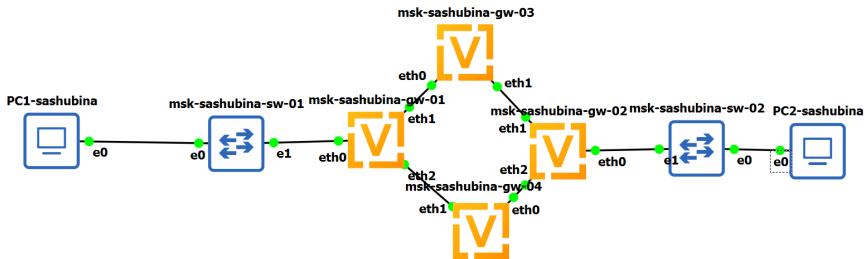


Рис. 129: топология

2. Заполнила таблицы адресации для всех устройств в сети

gw-01	eth0	10.10.1.97/27	n/a	PC1	
gw-01	eth0	2001:db8:1:1::1/64	n/a	PC1	
gw-01	eth1	10.10.1.5/30	n/a	gw-03	
gw-01	eth1	2001:db8:1:2::1/64	n/a	gw-03	
gw-01	eth2	10.10.1.33/30	n/a	gw-04	
gw-01	eth2	2001:db8:1:5::1/64	n/a	gw-04	
gw-02	eth0	10.10.1.65/28	n/a	PC2	
gw-02	eth0	2001:db8:1:6::1/64	n/a	PC2	
gw-02	eth1	10.10.1.18/30	n/a	gw-03	
gw-02	eth1	2001:db8:1:4::2/64	n/a	gw-03	
gw-02	eth2	10.10.1.9/30	n/a	gw-04	
gw-02	eth2	2001:db8:1:3::1/64	n/a	gw-04	
gw-03	eth0	10.10.1.8/30	n/a	gw-01	
gw-03	eth0	2001:db8:1:2::2/64	n/a	gw-01	
gw-03	eth1	10.10.1.17/30	n/a	gw-02	
gw-03	eth1	2001:db8:1:4::1/64	n/a	gw-02	
gw-04	eth0	10.10.1.10/30	n/a	gw-02	
gw-04	eth0	2001:db8:1:3::2/64	n/a	gw-02	
gw-04	eth1	10.10.1.34/30	n/a	gw-01	
gw-04	eth1	2001:db8:1:5::2/64	n/a	gw-01	
PC1	NIC	10.10.1.98/27	10.10.1.97	gw-01	
PC1	NIC	2001:db8:1:1::2/64	n/a	gw-01	
PC2	NIC	10.10.1.66/28	10.10.1.65	gw-02	

Рис. 130: таблица адресации

3. Задала адресацию PC1,PC2

```
VPCS> ip 10.10.1.98/27 10.10.1.97
Checking for duplicate address...
VPCS : 10.10.1.98 255.255.255.224 gateway 10.10.1.97

VPCS> ip 2001:db8:1:1::2/64
PC1 : 2001:db8:1:1::2/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME          : VPCS[1]
IP/MASK       : 10.10.1.98/27
GATEWAY       : 10.10.1.97
DNS           :
MAC           : 00:50:79:66:68:00
LPORT        : 10018
RHOST:PORT    : 127.0.0.1:10019
MTU           : 1500

VPCS> show ipv6

NAME          : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 2001:db8:1:1::2/64
DNS             :
ROUTER LINK-LAYER :
MAC             : 00:50:79:66:68:00
LPORT          : 10018
RHOST:PORT      : 127.0.0.1:10019
MTU             : 1500

VPCS> █
```

Рис. 131: PC1

3. Задала адресацию PC1,PC2

```
VPCS> ip 10.10.1.66/28 10.10.1.65
Checking for duplicate address...
VPCS : 10.10.1.66 255.255.255.240 gateway 10.10.1.65

VPCS> ip 2001:db8:1:6::2/64
PC1 : 2001:db8:1:6::2/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME          : VPCS[1]
IP/MASK       : 10.10.1.66/28
GATEWAY       : 10.10.1.65
DNS           :
MAC           : 00:50:79:66:68:01
LPORT        : 10016
RHOST:PORT    : 127.0.0.1:10017
MTU           : 1500

VPCS> show ipv6

NAME          : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 2001:db8:1:6::2/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:01
LPORT          : 10016
RHOST:PORT     : 127.0.0.1:10017
MTU            : 1500

VPCS> █
```

Рис. 132: PC2

4. Задала хост имя

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-sashubina-gw-01
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-sashubina-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N]
```

Рис. 133: gw-01

4. Задала хост имя

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-sashubina-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-sashubina-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] █
```

Рис. 134: gw-02

4. Задала хост имя

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-sashubina-gw-03
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-sashubina-gw-03
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N]
```

Рис. 135: gw-03

4. Задала хост имя

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-sashubina-gw-04
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-sashubina-gw-04
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] █
```

Рис. 136: gw-04

5. Далее я назначила адресацию маршрутизаторам

```
vyos@sak-sashubina-gw-01:~$ configure
[edit]
vyos@sak-sashubina-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@sak-sashubina-gw-01# set interfaces ethernet eth0 address 10.10.1.97/27
[edit]
vyos@sak-sashubina-gw-01# set interfaces ethernet eth1 address 10.10.1.5/30
[edit]
vyos@sak-sashubina-gw-01# set interfaces ethernet eth2 address 10.10.1.33/30
[edit]
vyos@sak-sashubina-gw-01# set interfaces ethernet eth0 address 2001:db8:1::1::1/64
[edit]
vyos@sak-sashubina-gw-01# set service router-advert interface eth0 prefix 2001:db8:1::1::1/64
[edit]
vyos@sak-sashubina-gw-01# set interfaces ethernet eth1 address 2001:db8:1:2::1/64
[edit]
vyos@sak-sashubina-gw-01# set interfaces ethernet eth2 address 2001:db8:1:5::1/64
[edit]
vyos@sak-sashubina-gw-01# compare
[edit interfaces ethernet eth0]
+address dhcp
+address 10.10.1.97/27
+address 2001:db8:1::1::1/64
[edit interfaces ethernet eth1]
+address 10.10.1.5/30
+address 2001:db8:1:2::1/64
[edit interfaces ethernet eth2]
+address 10.10.1.33/30
+address 2001:db8:1:5::1/64
[edit service]
+router-advert {
+  interface eth0 {
+    prefix 2001:db8:1::1::1/64
+  }
+}
[edit]
vyos@sak-sashubina-gw-01# save
Warning: you have uncommitted changes that will not be saved.
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@sak-sashubina-gw-01#
```

Рис. 137: gw-01

5. Далее я назначила адресацию маршрутизаторам

```
vyos@nak-sashubina-gw-02:~$ configure
[edit]
vyos@nak-sashubina-gw-02# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@nak-sashubina-gw-02# set interfaces ethernet eth0 address 10.10.1.65/28
[edit]
vyos@nak-sashubina-gw-02# set interfaces ethernet eth1 address 10.10.1.18/30
[edit]
vyos@nak-sashubina-gw-02# set interfaces ethernet eth2 address 10.10.1.9/30
[edit]
vyos@nak-sashubina-gw-02# set interfaces ethernet eth0 address 2001:db8:1:6::1/64
[edit]
vyos@nak-sashubina-gw-02# set service router-advert interface eth0 prefix 2001:db8:1:6::/64
[edit]
vyos@nak-sashubina-gw-02# set interfaces ethernet eth1 address 2001:db8:1:4::2/64
[edit]
vyos@nak-sashubina-gw-02# set interfaces ethernet eth2 address 2001:db8:1:3::1/64
[edit]
vyos@nak-sashubina-gw-02# compare
[edit interfaces ethernet eth0]
+address dhcp
+address 10.10.1.65/28
+address 2001:db8:1:6::1/64
[edit interfaces ethernet eth1]
+address 10.10.1.18/30
+address 2001:db8:1:4::2/64
[edit interfaces ethernet eth2]
+address 10.10.1.9/30
+address 2001:db8:1:3::1/64
[edit service]
+router-advert {
+  interface eth0 {
+    prefix 2001:db8:1:6::/64
+  }
+}
[edit]
vyos@nak-sashubina-gw-02# save
Warning: you have uncommitted changes that will not be saved.
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@nak-sashubina-gw-02#
```

Рис. 138: gw-02

5. Далее я назначила адресацию маршрутизаторам

```
vyos@msk-sashubina-gw-03:~$ configure
[edit]
vyos@msk-sashubina-gw-03# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-sashubina-gw-03# set interfaces ethernet eth0 address 10.10.1.6/30
[edit]
vyos@msk-sashubina-gw-03# set interfaces ethernet eth1 address 10.10.1.17/30
[edit]
vyos@msk-sashubina-gw-03# set interfaces ethernet eth0 address 2001:db8:1:2::2/64
[edit]
vyos@msk-sashubina-gw-03# set interfaces ethernet eth1 address 2001:db8:1:4::1/64
[edit]
vyos@msk-sashubina-gw-03# compare
[edit interfaces ethernet eth0]
- address dhcp
+ address 10.10.1.6/30
+ address 2001:db8:1:2::2/64
[edit interfaces ethernet eth1]
+ address 10.10.1.17/30
+ address 2001:db8:1:4::1/64
[edit]
vyos@msk-sashubina-gw-03# save
Warning: you have uncommitted changes that will not be saved.

Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-03#
```

Рис. 139: gw-03

5. Далее я назначила адресацию маршрутизаторам

```
vyos@msk-sashubina-gw-04:~$ configure
[edit]
vyos@msk-sashubina-gw-04# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-sashubina-gw-04# set interfaces ethernet eth0 address 10.10.1.10/30
[edit]
vyos@msk-sashubina-gw-04# set interfaces ethernet eth1 address 10.10.1.34/30
[edit]
vyos@msk-sashubina-gw-04# set interfaces ethernet eth0 address 2001:db8:1:3::2/64
[edit]
vyos@msk-sashubina-gw-04# set interfaces ethernet eth1 address 2001:db8:1:5::2/64
[edit]
vyos@msk-sashubina-gw-04# compare
[edit interfaces ethernet eth0]
- address dhcp
+ address 10.10.1.10/30
+ address 2001:db8:1:3::2/64
[edit interfaces ethernet eth1]
+ address 10.10.1.34/30
+ address 2001:db8:1:5::2/64
[edit]
vyos@msk-sashubina-gw-04# save
Warning: you have uncommitted changes that will not be saved.

Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-04#
```

Рис. 140: gw-04

6. Настроим маршрутизацию по протоколу RIP и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-01# set protocols rip interface eth0
[edit]
vyos@msk-sashubina-gw-01# set protocols rip interface eth1
[edit]
vyos@msk-sashubina-gw-01# set protocols rip interface eth2
[edit]
vyos@msk-sashubina-gw-01# commit
[edit]
vyos@msk-sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-01#
```

Рис. 141: gw-01

6. Настроим маршрутизацию по протоколу RIP и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-01:~$ show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network      Next Hop      Metric From      Tag Time
C(i) 10.10.1.4/30   0.0.0.0       1 self          0
C(i) 10.10.1.32/30  0.0.0.0       1 self          0
C(i) 10.10.1.96/27  0.0.0.0       1 self          0
vyos@msk-sashubina-gw-01:~$ show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 11 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive any version
    Interface      Send Recv  Key-chain
    eth0           2    1 2
    eth1           2    1 2
    eth2           2    1 2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
  Distance: (default is 120)
vyos@msk-sashubina-gw-01:~$
```

Рис. 142: gw-01

7. Пропингуем оконечное устройство

```
VPCS> trace 10.10.1.66
traceroute to 10.10.1.66, 8 hops max
 1 10.10.1.66      0.001 ms

VPCS> ping 10.10.1.66

10.10.1.66 icmp_seq=1 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=2 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=3 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=4 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=5 ttl=64 time=0.001 ms

VPCS> █
```

Рис. 143: PC2

8. Просмотрим захваченный трафик

```
0100 .... = Version: 4
..... 0101 = Header Length: 10 bytes (5)
> Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  Total Length: 132
  Identification: 0xdb1a (56090)
> 010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 1
  Protocol: UDP (17)
  Header Checksum: 0xb21a [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.10.1.97
  Destination Address: 224.0.0.9
  [Stream index: 4]
✓ User Datagram Protocol, Src Port: 520, Dst Port: 520
  Source Port: 520
  Destination Port: 520
  Length: 112
  Checksum: 0xd5e1 [unverified]
  [Checksum status: Unverified]
  [Stream index: 1]
  [Stream Packet Number: 19]
> [Timestamps]
  UDP payload (104 bytes)
✓ Routing Information Protocol
  Command: Response (2)
  Version: RIPv2 (2)
  ✓ IP Address: 10.10.1.4, Metric: 1
    Address Family: IP (2)
    Route Tag: 0
    IP Address: 10.10.1.4
    Netmask: 255.255.255.252
    Next Hop: 0.0.0.0
    Metric: 1
  > IP Address: 10.10.1.0, Metric: 2
  > IP Address: 10.10.1.16, Metric: 2
  > IP Address: 10.10.1.32, Metric: 1
  > IP Address: 10.10.1.64, Metric: 3
```

Рис. 144: ваиршарк

9. Настроим маршрутизацию по протоколу RIPng и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-02# set protocols ripng interface eth0
[edit]
vyos@msk-sashubina-gw-02# set protocols ripng interface eth1
[edit]
vyos@msk-sashubina-gw-02# set protocols ripng interface eth2
[edit]
vyos@msk-sashubina-gw-02# commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02#
```

Рис. 145: gw-02

9. Настроим маршрутизацию по протоколу RIPng и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-01:~$ show ipv6 ripng status
Routing Protocol is "RIPng"
  Sending updates every 30 seconds with +/-50%, next due in 5 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 1, receive version 1
    Interface          Send  Recv
    eth0                1     1
    eth1                1     1
    eth2                1     1
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway             BadPackets BadRoutes Distance Last Update
vyos@msk-sashubina-gw-0 :~$ show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface, (a/S) - aggregated/Suppressed

   Network      Next Hop          Via      Metric Tag Time
C(i) 2001:db8:1:3::/64
      ::
      self       1    0
C(i) 2001:db8:1:4::/64
      ::
      self       1    0
C(i) 2001:db8:1:6::/64
      ::
      self       1    0
vyos@msk-sashubina-gw-02:~$
```

Рис. 146: gw-01

10. Пропингуем оконечное устройство

```
VPCS> ping 2001:db8:1:6::2/64

2001:db8:1:6::2 icmp_seq=1 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=2 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=3 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=4 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=5 ttl=64 time=0.001 ms

VPCS> trace 2001:db8:1:6::2/64

trace to 2001:db8:1:6::2, 64 hops max
 1 2001:db8:1:6::1    14.239 ms   5.903 ms   10.296 ms
 2 2001:db8:1:6::1    7.660 ms   5.153 ms   24.548 ms
 3 2001:db8:1:6::1    9.754 ms   20.501 ms  13.540 ms
 4 2001:db8:1:6::1   30.441 ms   4.562 ms   13.206 ms
```

Рис. 147: ping

11. Настроим маршрутизацию по протоколу OSPF и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-02:~$ configure
[edit]
vyos@msk-sashubina-gw-02# set protocols ospf area 0 network 10.10.1.64/28
[edit]
vyos@msk-sashubina-gw-02# set protocols ospf area 0 network 10.10.1.17/30

  10.10.1.17/30 is an IPv4 host address, not a network address. Did you mean 10.
10.1.16/30?

Value validation failed
Set failed

[edit]
vyos@msk-sashubina-gw-02# set protocols ospf area 0 network 10.10.1.16/30
[edit]
vyos@msk-sashubina-gw-02# set protocols ospf area 0 network 10.10.1.8/30
[edit]
vyos@msk-sashubina-gw-02# commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02#
```

Рис. 148: gw-02

```
vyos@msk-sashubina-gw-01:~$ show ipv6 ospfv3 route
*N IA 2001:db8:1:3::/64 ::
```

eth2 00:00:46

12. Пропингуем оконечное устройство

```
VPCS> ping 10.10.1.66

10.10.1.66 icmp_seq=1 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=2 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=3 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=4 ttl=64 time=0.001 ms
10.10.1.66 icmp_seq=5 ttl=64 time=0.001 ms

VPCS> trace 10.10.1.66
traceroute to 10.10.1.66, 8 hops max
 1 10.10.1.66      0.001 ms

VPCS> █
```

Рис. 150: пинг

13. Захваченный трафик

```
> Frame 484: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface --, Id 0
▼ Ethernet II, Src: 0c:97:0f:bc:00:00 (0c:97:0f:bc:00:00), Dst: IPvencast_05 (01:00:5e:00:00:05)
  > Destination: IPvencast_05 (01:00:5e:00:00:05)
  > Source: 0c:97:0f:bc:00:00 (0c:97:0f:bc:00:00)
  Type: IPv4 (0x0800)
  [Stream index: 11]
▼ Internet Protocol Version 4, Src: 10.10.1.97, Dst: 224.0.0.5
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  Total Length: 64
  Identification: 0xaf59 (44889)
  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 1
  Protocol: OSPF ISOP (89)
  Header Checksum: 0x1dc [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.10.1.97
  Destination Address: 224.0.0.5
  [Stream index: 7]
▼ Open Shortest Path First
  ▼ OSPF Header
    Version: 2
    Message Type: Hello Packet (1)
    Packet Length: 44
    Source OSPF Router: 10.10.1.97
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0x4e48 [correct]
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  ▼ OSPF Hello Packet
    Network Mask: 255.255.255.224
    Hello Interval [sec]: 10
    > Options: 0x02, (E) External Routing
    Router Priority: 1
    Router Dead Interval [sec]: 40
    Designated Router: 10.10.1.97
    Backup Designated Router: 0.0.0.0
```

Рис. 151: ваишрак

14. Настроим маршрутизацию по протоколу OSPFv3 и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-02:~$ configure
[edit]
vyos@msk-sashubina-gw-02# set protocols ospfv3 parameters router-id 2.2.2.2
[edit]
vyos@msk-sashubina-gw-02# set protocols ospfv3 area 0 interface eth0
[edit]
vyos@msk-sashubina-gw-02# set protocols ospfv3 area 0 interface eth1
[edit]
vyos@msk-sashubina-gw-02# set protocols ospfv3 area 0 interface eth2
[edit]
vyos@msk-sashubina-gw-02# commit
[edit]
vyos@msk-sashubina-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-sashubina-gw-02#
```

Рис. 152: gw-02

14. Настроим маршрутизацию по протоколу OSPFv3 и посмотрим метрики протокола

```
vyos@msk-sashubina-gw-01:~$ show ip ospf route
===== OSPF network routing table =====
N      10.10.1.8/30          [100] area: 0.0.0.0
                        directly attached to eth2
N      10.10.1.16/30        [100] area: 0.0.0.0
                        directly attached to eth1
N      10.10.1.64/28        [100] area: 0.0.0.0
                        directly attached to eth0

===== OSPF router routing table =====

===== OSPF external routing table =====
```

Рис. 153: gw-01

15. пропингуем окончечное устройство

```
VPCS> ping 2001:db8:1:6::2/64

2001:db8:1:6::2 icmp_seq=1 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=2 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=3 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=4 ttl=64 time=0.001 ms
2001:db8:1:6::2 icmp_seq=5 ttl=64 time=0.001 ms

VPCS> trace 2001:db8:1:6::2/64

trace to 2001:db8:1:6::2, 64 hops max
 1 2001:db8:1:6::1    14.239 ms   5.903 ms   10.296 ms
 2 2001:db8:1:6::1    7.660 ms   5.153 ms   24.548 ms
 3 2001:db8:1:6::1    9.754 ms   20.501 ms   13.540 ms
 4 2001:db8:1:6::1   30.441 ms   4.562 ms   13.206 ms
```

Рис. 154: пинг

3. Захват трафика

```
> Frame 557: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface -, id 0
✓ Ethernet II, Src: 0c:8e:91:83:00:00 (0c:8e:91:83:00:00), Dst: IPv6mcast_05 (33:33:00:00:00:05)
  > Destination: IPv6mcast_05 (33:33:00:00:00:05)
  > Source: 0c:8e:91:83:00:00 (0c:8e:91:83:00:00)
  Type: IPv6 (0x86dd)
  [Stream index: 12]
✓ Internet Protocol Version 6, Src: fe80::e8e:91ff:fe83:0, Dst: ff02::5
  0110 .... = Version: 6
  > .... 1100 0000 .... = Traffic Class: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  .... 0100 1011 1011 1010 1000 = Flow Label: 0x4bba8
  Payload Length: 36
  Next Header: OSPF TGP (89)
  Hop Limit: 1
  > Source Address: fe80::e8e:91ff:fe83:0
  > Destination Address: ff02::5
  [Source SLAAC MAC: 0c:8e:91:83:00:00 (0c:8e:91:83:00:00)]
  [Stream index: 0]
✓ Open Shortest Path First
  ✖ OSPF Header
    Version: 3
    Message Type: Hello Packet (1)
    Packet Length: 36
    Source OSPF Router: 2.2.2.2
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0x5674 [correct]
    Instance ID: IPv6 unicast AF (0)
    Reserved: 00
  ✖ OSPF Hello Packet
    Interface ID: 2
    Router Priority: 1
    > Options: 0x00013, R, E, V6
    Hello Interval [sec]: 10
    Router Dead Interval [sec]: 40
    Designated Router: 2.2.2.2
    Backup Designated Router: 0.0.0.0
```

Рис. 155: виршарк

Я изучила принципы маршрутизации в IPv4- IPv6-сетях и принципы настройки сетевого оборудования